

# 55 HMI DEVICE COMMUNICATION

The HMI device communication function allows to communicate with external HMI devices etc. that are connected via RS-232-C or Ethernet via Modbus/TCP or OPC UA. Various robot data, such as DI, DO, Register and so on, can be read or written from the HMI device etc. by using this function.

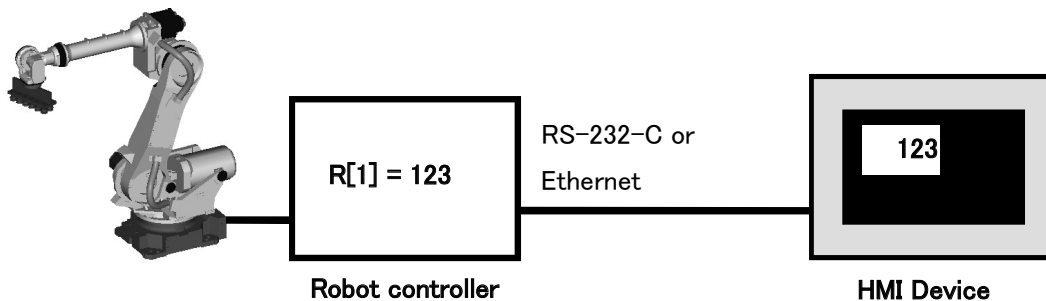
This function operates as both:

*Modbus slave, communicating with HMI device etc. that is the Modbus master*

And,

*OPC UA Server, communicating with OPC UA Clients. (Robot can't become OPC UA client.)*

HMI device can design the screen by putting lumps, switches and so on. The assignment of robot data to the lumps, switches and so on can be defined by user. For example, if robot register is assigned to the display panel on HMI device, the register value can be displayed on the screen of HMI device.



This function is standard when "Standard setting (R651)" is selected, but the following version conditions must be met in order to use each function. When "North America Setting (R650)" is selected, "HMI device (R553)" option is needed to use this function.

- Modbus slave function is supported from version 7DC2 series P07 and all versions after 7DC2 series. "HMI device (R553)" doesn't support Modbus slave function in 7DC1 series or before.
- OPC UA server function is supported from 7DF1 series P19 or after and all versions after 7DF1 series.
- OPC UA server on ROBOGUIDE is supported from 7DF1 series P24 or after, 7DF3 series P04 or after, and all versions after 7DF3 series.

The response time of this function may 3-18ms usually but may take 100ms or more and is not guaranteed. It depends on the robot controller load.

If you want to get real time current position, use "HIGH SPEED POSITION OUTPUT" in section 68.

OPC UA server function supports Publisher and MonitoredItem so client can use subscriber.

Minimum data acquiring interval of MonitoredItem is 100ms.

In OPC UA server function, OPC UA client will read/write large array data so we recommend to use the client soft which support index range, which enables to read/write specific range only.

**⚠ WARNING**

This function checks the received data and performs the proper process only when the robot controller receives the data sent from HMI device etc. This function cannot detect the situation that the communication from HMI device etc. is stopped accidentally. Therefore, this function must not be used to transfer the safety data. For example, when HMI device etc. sends STOP request to robot, there is no guarantee that the robot is stopped by the request. It is very dangerous to make safety system by using this function.

**NOTE**

When this function is used to communicate with HMI device etc., please test the whole robot system including the HMI device etc.

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**Connection tested HMI devices****Schneider Electric Japan Holdings Ltd. GP4000 Series**

Set followings to the “Manufacturer” and “Series”.

RS-232-C : "Manufacturer: Modbus-IDA", "Series: MODBUS RTU SIO master"

Ethernet: "Manufacturer: Modbus-IDA", "Series: MODBUS TCP master"

**KEYENCE CORPORATION VT3 Series**

Set followings to the “Manufacturer” and “Model”.

RS-232-C: "Manufacturer: MODBUS protocol", "Model: RTU mode (1:N)"

Ethernet: "Manufacturer: MODBUS protocol", "Model: MODBUS/TCP(Ethernet)"

## 55.1 CONNECTION OF HMI DEVICE

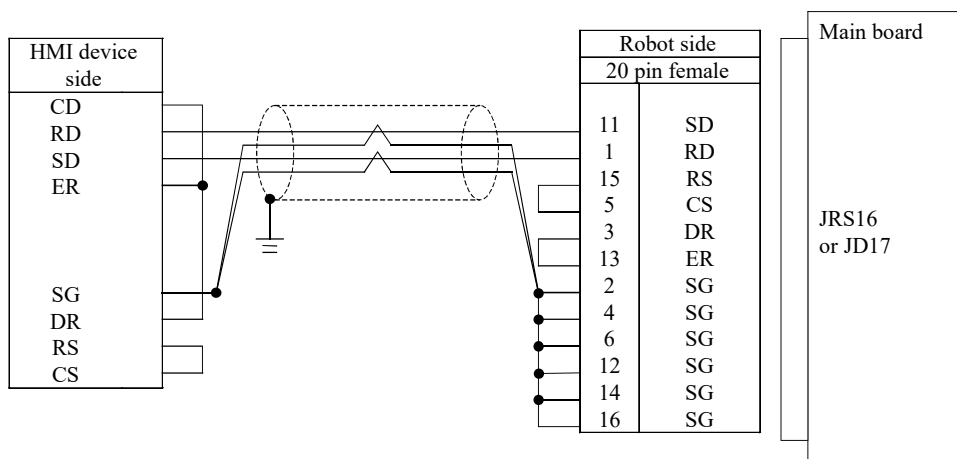
### 55.1.1 Modbus RS-232-C Connection

**Cable**

The following diagram shows the connection of RS-232-C. Please refer to the manual of the HMI device about the connection of HMI device side.

**A-cabinet**

The cable of the ROBOT CONTROLLER side is connected to JRS16 (Port 1) or JD17 (Port 2) connector on the main board.

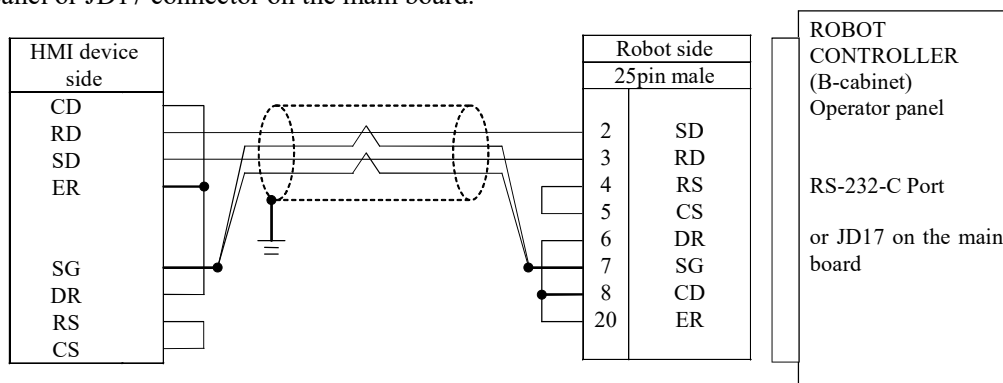


Specification of the connectors and the cable are as following:

- Connector PCR-E20FS (Honda Tsushin Kogyo)
- Housing PCR-V20LA (Honda Tsushin Kogyo), or compatible connector
- Shielding quality Entirely shielded twisted pair cable is recommended.  
For protection against the noise, the twisted pair wire should be used for pairs of RD and SG, SD and SG.

## B-cabinet

The D-Sub 25pin is connected to the RS-232-C port in front side of ROBOT CONTROLLER operator panel or JD17 connector on the main board.

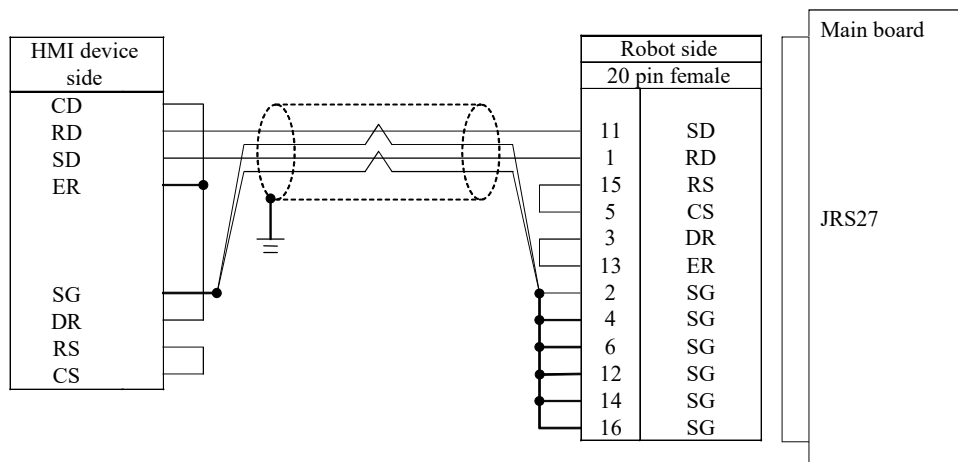


Specification of the connectors and the cable are as following (Operator panel):

- Connector D-Sub25pin male DBM-25P (ANSI/EIA-232)
- Housing DB-C2-J9 (ANSI/EIA-232)
- Shielding quality Entirely shielded twisted pair cable is recommended.  
For protection against the noise, the twisted pair wire should be used for pairs of RD and SG, SD and SG.

**R-30iB (Mate / Compact / Mini) (Plus)**

The cable of the ROBOT CONTROLLER side is connected to JRS27 connector on the main board.



Specification of the connectors are as following:

- Connector PCR-E20FS (Honda Tsushin Kogyo)
- Housing PCR-V20LA (Honda Tsushin Kogyo), or compatible connector
- Shielding quality Entirely shielded twisted pair cable is recommended.  
For protection against the noise, the twisted pair wire should be used for pairs of RD and SG, SD and SG.

**Setting of Serial Port**

To use HMI device communication via RS-232-C, it is necessary to set "HMI dev(MODBUS)" in Port Init menu on Robot controller.

- 1 Press the [MENU] key on teach pendant, and select "6 SETUP".
- 2 Press the F1[TYPE] key, and select "Port Init".

| Port Init            |                    | 1/2 |
|----------------------|--------------------|-----|
| Connector            | Comment            |     |
| 1 JRS16 RS-232-C P2: | [Maintenance Cons] |     |
| 2 JD17 RS-232-C P3:  | [KCL/CRT]          |     |
|                      |                    | 1   |
| [ TYPE ]             | DETAIL             |     |

- 3 Set cursor on "JRS16 RS-232-C P2:" line, and press the F3(DETAIL) key. The detail menu of "JRS16 RS-232-C P2:" is displayed.

| Port Init              |                    | 1/5      |
|------------------------|--------------------|----------|
| JRS16 RS-232-C P2:     |                    |          |
| 1 Device               | [Maintenance Cons] |          |
| 2 Speed                | [9600]             |          |
| 3 Parity bit           | [None]             |          |
| 4 Stop bit             | [1bit]             |          |
| 5 Time out value (sec) | [ 0]               |          |
| [ TYPE ]               | LIST               | [CHOICE] |

- 4 Set cursor on "Device" line, and press the F4[CHOICE], and select "HMI dev(MODBUS)" from the list. There are "HMI device" and "HMI dev(MODBUS)" in the list. It is necessary to set "HMI

dev(MODBUS)" for Modbus communication. It is not possible to use "HMI device" and "HMI dev(MODBUS)" at the same time.

| Port Init              |                   |
|------------------------|-------------------|
| JRS16 RS-232-C P2:     | 1/5               |
| 1 Device               | [HMI dev(MODBUS)] |
| 2 Speed                | [19200]           |
| 3 Parity bit           | [Even ]           |
| 4 Stop bit             | [1bit ]           |
| 5 Time out value (sec) | [ 0]              |

| [ TYPE ] | LIST | [CHOICE] |
|----------|------|----------|
|          |      |          |

- 5 Setup "Speed", "Parity bit" and "Stop bit" as same with the HMI device.

### Setup slave address

The slave address of this function is set in the system variable \$SNPX\_PARAM.\$MODBUS\_ADR. The default value is 1. The slave address of Modbus RTU can be changed by this system variable. This function responds to the request for the specified slave address only. The request for the slave address 0 is processed as broadcast.

## 55.1.2 Modbus/TCP Slave Ethernet Connection

### Setting of TCP/IP

To use HMI device communication via Modbus/TCP Slave Ethernet, it is necessary to setup TCP/IP on Robot controller in advance.

Details on the Ethernet interface and TCP/IP configuration can be found in "SETTING UP TCP/IP" section in the *Internet Options Setup and Operations Manual*.

### Setting of Modbus/TCP Slave Ethernet connection

Modbus/TCP Slave Ethernet connection is disable by default. To use Modbus/TCP Slave Ethernet connection, please set the number of HMI devices etc. that are connected at the same time to the system variable \$SNPX\_PARAM.\$NUM\_MODBUS (Default is 0). When a Modbus/TCP Slave Ethernet connection is requested from HMI devices etc. which would exceed the setting value, the HMI device etc. whose latest communication is the oldest is disconnected automatically.

### Unit ID

Robot controller does not use Unit ID in Modbus TCP communication data. Robot controller responds all requests from an HMI device etc. regardless of Unit ID.

### Port number

The port number of this function is set in the system variable \$SNPX\_PARAM.\$MODBUS\_PORT. The default value is 502. The port number of this function can be changed by this system variable.

### Modbus TCP Server (R800)

Modbus TCP Server (R800) option is independent from this function. But when this option is loaded, if \$SNPX\_PARAM.\$NUM\_MODBUS is set to 1 or greater value, "PRIO-090 SNPX communication error", "HRTL-048 Address already in use" occurs and Modbus/TCP Slave Ethernet communication of this function does not work. It is because this function and Modbus TCP Server use the same port number (502).

By changing the port number of this function, both this function and Modbus TCP Server can be used at the same time.

## 55.1.3 OPC UA Server Ethernet Connection

OPC UA is multi-purpose and flexible industrial communication protocol which enables data exchange between multi-vendor products and different operating systems. This function enables the FANUC robot to connect with OPC UA clients and enables various robot data (such as DI, DO, Registers, and so on) to be read and written. This function works as an OPC UA server and supports communication with external OPC UA clients over Ethernet.

The information model (data model) used for sending and receiving data is based on the Modbus data model of the “HMI DEVICE COMMUNICATION” function.

### Limitation

The following limitations exist when using the FANUC robot OPC UA Server.

- Maximum of ten (10) OPC UA clients can communicate simultaneously.
- This function can be used only with CD38A and CD38B RJ45 Ethernet port connections on the robot controller. This function cannot be used with the CD38C RJ45 Ethernet port connection.
- Some OPC UA client can set subscription and monitored item. By setting this, the client can get the item value when updated. For robot controller, up to 2 subscriptions can be registered per 1 connection. (It also supports the Client/Server method.)
- For robot controller, up to 50 items can be monitored per 1 subscription but minimum data acquiring interval (Minimum notification interval to client) for the monitored item is increased by 100ms per 1 item like 100ms for 1<sup>st</sup> item, 200ms for 2<sup>nd</sup> item, 300ms for 3<sup>rd</sup> item. This does NOT mean the 4<sup>th</sup> item would be updated only every 1 second (= 100ms + 200ms + 300ms + 400ms). 4<sup>th</sup> item can be updated every 400ms as minimum. Normal read / write (Data access(DA)) doesn't have such limitations.

### Setting of TCP/IP

To use HMI device communication via the FANUC robot OPC-UA Server, it is necessary to setup TCP/IP on the Robot controller in advance. See "2.4 SETTING UP TCP/IP" section in the *Ethernet Function OPERATOR'S MANUAL (B-82974EN)* or "2.5 SETTING UP TCP/IP" section in the *Internet Options Setup and Operations Manual*.

The FANUC robot OPC-UA Server can be accessed from an external OPC-UA client by entering the robot's OPC UA server URL in the following format:

`opc.tcp://robot IP address:4880/FANUC/NanoUaServer`

Example:

`opc.tcp://172.22.194.222:4880/FANUC/NanoUaServer`

The OPC UA client can connect with the robot OPC UA server after setting the above URL in the OPC UA client. Please see the “55.2 MODBUS DATA MODEL” sections for details about assigning and accessing robot data.

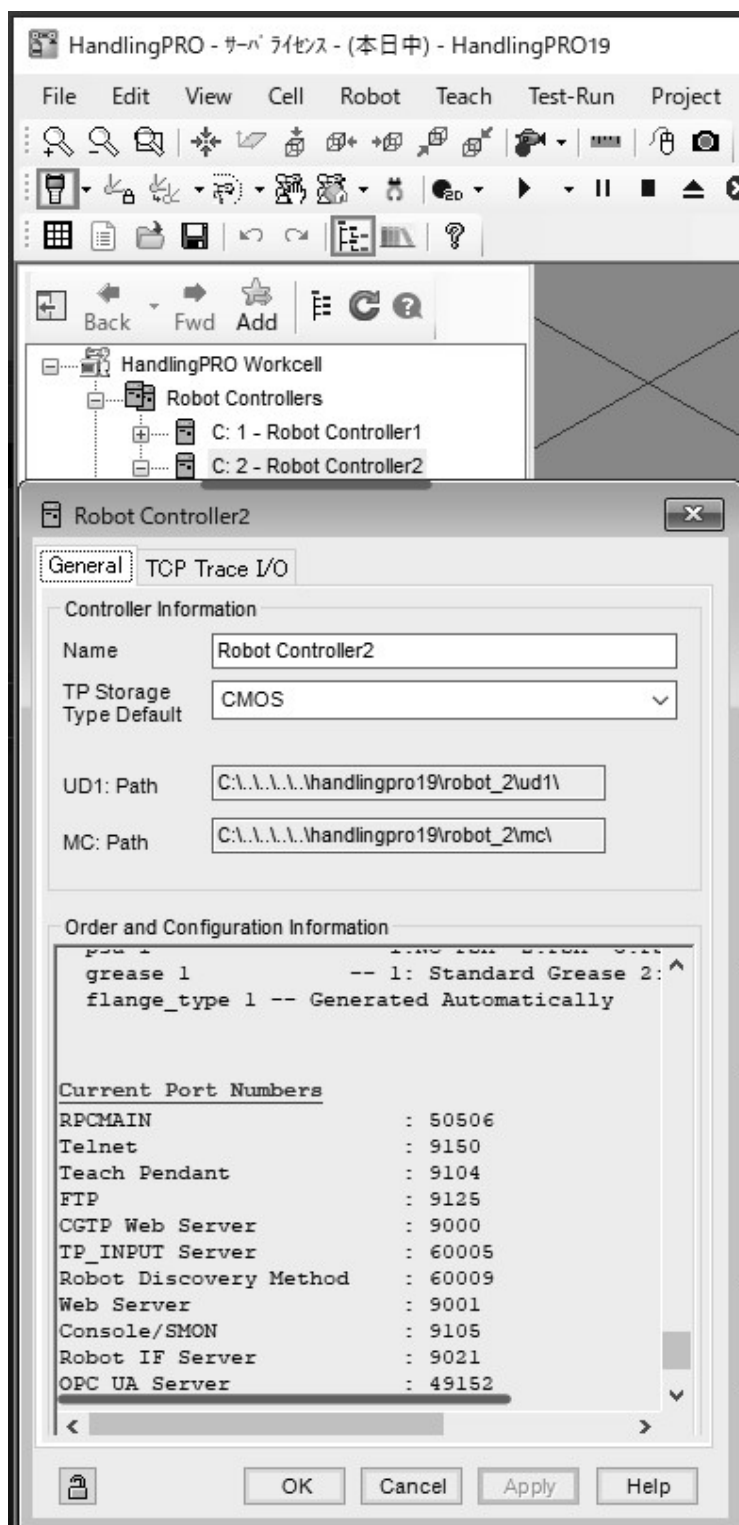
Authentication, user name and password aren't supported.

For ROBOGUIDE, set the following format URL to OPC UA client.

`opc.tcp://IP address for PC where ROBOGUIDE is running:4880/FANUC/NanoUaServer`

4880 in the URL should be changed to the number displayed at “OPC UA Server” at the bottom of the window, which is displayed after double clicking the connecting robot controller, in the following figure.

ROBOGUIDE supports OPC UA server function from 7DF1(V9.10) series version 24 or from 7DF3(V9.30) series version 04 or all versions after 7DF3(V9.30) series.



## 55.2 MODBUS DATA MODEL

### 55.2.1 Modbus data model

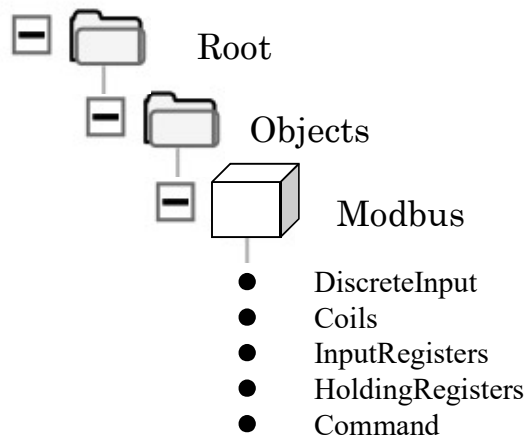
Modbus bases its data model on a series of tables.

| Table             | Object type | Type of    |
|-------------------|-------------|------------|
| Discrete input    | Single bit  | READ only  |
| Coils             | Single bit  | READ-WRITE |
| Input Registers   | 16-bit word | READ only  |
| Holding Registers | 16-bit word | READ-WRITE |

### 55.2.2 OPC UA Organization of Modbus data model

This section applies only when using OPC UA client for HMI device communication with the built-in FANUC robot OPC UA Server.

The “Modbus” object nodes can be accessed from the “Root/Object/Modbus” directory. As displayed below.



There are other folders just under root folder but is not used.

The following table describes node and data types of the five (5) leaf nodes under the “Modbus” object.

| Node             | NameSpace | Node ID | Data type | Data size |
|------------------|-----------|---------|-----------|-----------|
| DiscreteInput    | 1         | 301     | Boolean   | 1 byte    |
| Coils            |           | 302     | Boolean   | 1 byte    |
| InputRegisters   |           | 303     | UInt16    | 2 byte    |
| HoldingRegisters |           | 304     | Int16     | 2 byte    |
| Command          |           | 305     | String[]  | 128 byte  |

- Leaf nodes other than Command node are array. The index range should be supported on the OPC UA client to support partial read or write for the leaf nodes.
- For example, each X, Y, Z in position register cannot be assigned discrete Node ID.
- The value of the node is the latest value at the read time.

Please Note: In the following sections, the term *HMI Device etc.* can also refer to an OPC UA client.



### 55.2.3 Correspondence of Modbus Address to Robot Data

Each “Table” has address area 1 ~ 65536. This function defines the following correspondence of Modbus address to Robot data. The HMI device etc. can read or write the corresponded robot data by accessing Modbus address.

OPC UA clients can read or write the desired robot data by accessing the index which is “Modbus addresses - 1”.

| Table             | Address range | Corresponded robot data                                       |
|-------------------|---------------|---------------------------------------------------------------|
| Discrete input    | 1 ~ 10000     | Digital input DI[a]                                           |
|                   | 10001 ~ 20000 | Robot input RI[1-10000]                                       |
|                   | 20001 ~ 21000 | UOP input UI[1-10000]                                         |
|                   | 21001 ~ 21999 | UOP output UO[1-1000]                                         |
|                   | 22000 ~ 22999 | SOP input SI[0-999]                                           |
|                   | 23000 ~ 24000 | SOP output SO[0-1000]                                         |
|                   | 24001 ~ 25000 | Weld interface digital input WI[1-1000]                       |
|                   | 25001 ~ 26000 | Weld interface digital output WO[1-1000]                      |
|                   | 26001 ~ 27000 | Wire soldering detector input WSI[1-1000]                     |
|                   | 27001 ~ 28000 | Wire soldering detector output WSO[1-1000]                    |
|                   | 28001 ~ 65536 | Not used                                                      |
| Coils             | 1 ~ 10000     | Digital output DO[1-10000]                                    |
|                   | 10001 ~ 20000 | Robot output RO[1-10000]                                      |
|                   | 20001 ~ 30000 | Flag F[1-10000]                                               |
|                   | 30001 ~ 65536 | Not used                                                      |
| Input Registers   | 1 ~ 1000      | Group input GI[1-1000]                                        |
|                   | 1001 ~ 2000   | Group output GO[1-1000]                                       |
|                   | 2001 ~ 3000   | Analog input AI[1-1000]                                       |
|                   | 3001 ~ 4000   | Analog output AO[1-1000]                                      |
|                   | 4001 ~ 65536  | Not used                                                      |
| Holding Registers | 1 ~ 16384     | Robot data is assigned (assigned to register by default)      |
|                   | 16385 ~ 16448 | 55.4 DYNAMIC ASSIGNMENT TO HOLDING REGISTERS using this area. |
|                   | 16449 ~ 65536 | Not used                                                      |

When reading an address that is not used (or the corresponding robot data is not assigned) from an HMI device etc., the read value is always 0. When the address is written from an HMI device etc., the request is ignored. In these cases, this function does not return an error to the HMI device etc.

The address range 1 ~ 16384 of Holding Registers can be assigned to various Robot data. (→ Assignment of Holding Registers)

The HMI device etc. cannot write to the signals corresponded to Discrete input and Input Registers. It is possible to write these signals via Holding Registers by assigning WO, WSO, GO, AO in these signals to Holding Registers. (→ Assign I/O data and simulation status)

In this function, address range is specified as 1 ~ 65536. But some HMI devices etc. specify as 0000 ~ FFFF (start from 0 and specified as hexadecimal). In this case, please convert the specified address as subtraction of 1 and change to hexadecimal.

The maximum size that can be read or written at one time via Modbus communication is 507 words. The amount of communication is 2 packets per communication, and 6 packets when connecting and disconnecting. However, 2 packets are added for every 256 bytes of data.

## 55.2.4 Modbus Function Code

When using Modbus TCP communication (not OPC UA communication), this function supports the following function codes.

| Function code name       | Code |
|--------------------------|------|
| Read Coils               | 01h  |
| Read Discrete Inputs     | 02h  |
| Read Holding Registers   | 03h  |
| Read Input Register      | 04h  |
| Write Single Coil        | 05h  |
| Write Single Register    | 06h  |
| Write Multiple Coils     | 0Fh  |
| Write Multiple Registers | 10h  |

## 55.3 ASSIGNMENT OF HOLDING REGISTERS

The address range 1 ~ 16384 (0~16383 when OPC UA) of Holding Registers can be assigned to various Robot data. Each address has 2 bytes (16 bits) memory space. It is possible to assign 32 bit data to the continuous 2 addresses, and to assign string data to the continuous plural addresses.

The following robot data can be assigned to the Holding Registers.

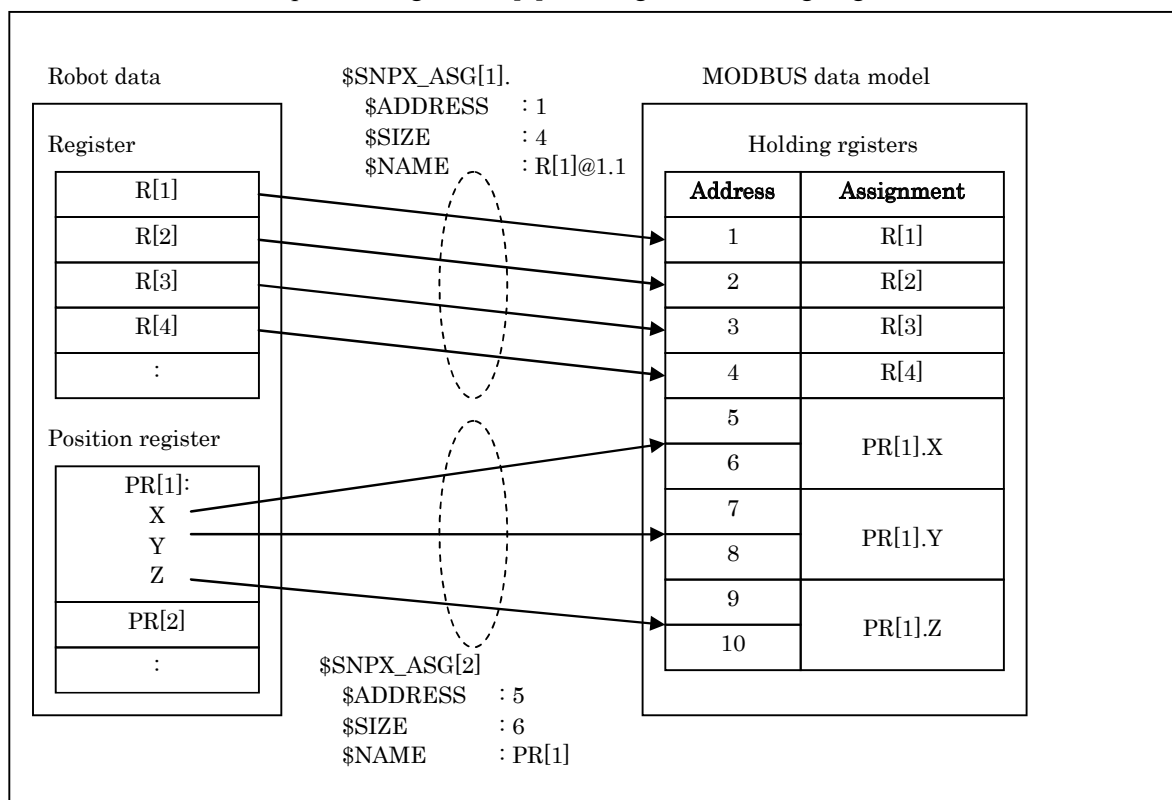
- Register data
- Position register data
- String register data
- Current position
- Alarm history
- Program execution status (Running program name and line number)
- System variable data
- Comment of Register, Position register, String register and I/O
- I/O data and simulation status
- Integrated PMC address data
- Symbol and comment of Integrated PMC address.

The assignment between Robot data and Holding Registers is defined by setting of \$SNPX\_ASG. The \$SNPX\_ASG is 80 arrays of \$SNPX\_ASG[1]~[80], every element has the following variables. Various robot data can be assigned to the Holding Registers by setting \$SNPX\_ASG.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign<br>Range: 1~16384                                                                                                                                                                                                                                                                                 |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                       |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>It specifies the robot data type and index by string. The meaning of this variable is different according to the assigned data type. Please refer to the description in each robot data.<br>Example:<br>R[1] : Register R[1]<br>PR[1] : Position register PR[1]<br>POS[1] : Current position of group 1 robot |

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$MULTIPLY             | <p>Meaning: Multiply for assigned data.</p> <p>It specifies the conversion type from assigned data to Holding Registers. The meaning of this variable is different according to the assigned data type. Please refer to the description in each robot data.</p> <p>Example: When register is assigned and the register value is 123.45, the value of the Holding Registers is the following.</p> <p>When \$MULTIPLY is 1, Holding Register is 123.</p> <p>When \$MULTIPLY is 10, Holding Register is 1235</p> <p>When \$MULTIPLY is 0.1, Holding Register is 12</p> |

The following figure shows a example that robot register R[1~4] are assigned to Holding Register address 1 ~ 4 and X,Y,Z of robot position register PR[1] are assigned to Holding Register address 5 ~ 10.



The default setting of \$SNPX\_ASG is the following.

| System variable          | Value    |
|--------------------------|----------|
| \$SNPX_ASG[1].\$ADDRESS  | 1        |
| \$SNPX_ASG[1].\$SIZE     | 10000    |
| \$SNPX_ASG[1].\$VAR_NAME | R[1]@1.1 |
| \$SNPX_ASG[1].\$MULTIPLY | 1        |

By the default setting, the assignment of Holding Registers is the following.

| Robot data    | Address of Holding Registers | Example          |
|---------------|------------------------------|------------------|
| Register R[x] | x                            | R[1] ⇔ Address 1 |

Note: Data type is 16 bits signed integer. Value is rounded off to no decimal place. Range of value is from -32768 to 32767. If value is out of the range, value cannot be read and write correctly.

### 55.3.1 Data type of Holding Registers

Data type of Holding Register is 16 bits integer. Data range of Holding Register is -32768 ~ 32767. The value of assigned robot data is converted as follows according to the data type of the assigned data.

| Assigned data                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 bits signed integer<br>Range: -32768 ~ 32767           | The Holding Register data is the same as the assigned data. (Big endian)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 32 bits signed integer<br>Range: -2147483648 ~ 2147483647 | The continued 2 Holding Registers express a 32 bits signed integer. The former address is lower 16 bits data, and the latter address is upper 16bits data.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Real                                                      | The continued 2 Holding Registers express a single precision binary real format of IEEE 754. The former address is lower 16 bits data, and the latter address is upper 16bits data.<br>In case of that real type robot data is assigned to Holding Registers, when \$MULTIPLY is 0, the robot data is assigned to Holding Registers as real type data. When \$MULTIPLY is not 0, the robot data is assigned to Holding Registers as 32 bits signed integer. In this case, the Holding Registers has the value that the robot data is multiplied by \$MULTIPLY and rounded off to no decimal place. |
| String                                                    | The continued Holding Registers express a string data. One Holding Register expresses 2 characters. Byte order of string data is specified by \$MULTIPLY as follows.<br>\$MULTIPLY = 1 (0 or more): Lower byte is the first character, upper byte is the second character.<br>\$MULTIPLY = -1 (less than 0): Upper byte is the first character, lower byte is the second character.<br>The last of the string data must be 0. When string data is written to Holding Registers, it is necessary to write 0 to the last of the string data.                                                         |



#### **WARNING**

If Integer data is written to the Holding Registers of Real data type, the assigned robot data may be set to the value that is too big or cannot be processed. And it may cause serious effect such as the robot moves to unexpected position because the assigned position register has abnormal position data. Please set \$MULTIPLY to the value of not 0 normally, and assign 32 bit signed integer data to Holding Registers.

### 55.3.2 Assign Robot Registers

\$SNPX\_ASG setting to assign Robot Registers to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                          |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                             |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One Register uses two Holding Registers.<br>(You can change the number of Holding Registers for one robot register by adding "@" in \$VAR_NAME)<br>Range: 1~16384 |

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$VAR_NAME             | <p>Meaning: String to specify robot data.<br/>Please set "R[1]" to assign Register R[1]. The number in [ ] is the index of register.</p> <p>Continued registers such as R[2]-R[5] can be assigned by one \$SNPX_ASG element. In this case, please set \$SIZE to 8 because the number of registers to assign is 4. And please set \$VAR_NAME "R[2]". The index 2 means the starting index of the continuous assignment.</p> <p>If "@1.1" is added just after the string, the number of Holding Registers for one register is changed from 2 to 1. In this case, accessing data size becomes 16 bits.<br/>Example: R[1]@1.1<br/>"R[1]" part means to assign robot registers from index 1.<br/>"@1.1" part means one register is accessed as 16 bits data.</p> |
| \$MULTIPLY             | <p>Meaning: Multiplier<br/>The value accessed by an HMI device etc. is multiplied by \$MULTIPLY.<br/>When \$MULTIPLY is 0, it means that an HMI device etc. can access Holding Registers as 32bits REAL data.<br/>When it is not 0, an HMI device etc. accesses Holding Registers as 32bits signed integer. And value is rounded off to no decimal place.<br/>Range: 0.0001~10000, 0<br/>Example: Register value is 123.45.<br/>When \$MULTIPLY is 1, Holding Register is 123<br/>When \$MULTIPLY is 10, Holding Register is 1235.<br/>When \$MULTIPLY is 0.1, Holding Register is 12.<br/>When \$MULTIPLY is 0, Holding Register is 123.45 of REAL</p>                                                                                                     |

For example, \$SNPX\_ASG is set as follows.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 2      | R[1]@1.1   | 1          |
| \$SNPX_ASG[2] | 3         | 4      | R[1]       | 100        |
| \$SNPX_ASG[3] | 7         | 4      | R[2]       | 0.1        |
| \$SNPX_ASG[4] | 11        | 2      | R[1]       | 0          |

In this case, address of Holding Registers are corresponded to robot registers as follows.

| Address | Assigned data                                    |
|---------|--------------------------------------------------|
| 1       | R[1] as 16bits signed integer.                   |
| 2       | R[2] as 16bits signed integer.                   |
| 3-4     | R[1] multiplied by 100 as 32bits signed integer. |
| 5-6     | R[2] multiplied by 100 as 32bits signed integer. |
| 7-8     | R[2] divided by 10 as 32bits signed integer.     |
| 9-10    | R[3] divided by 10 as 32bits signed integer.     |
| 11-12   | R[1] as 32bits REAL                              |

\$SNPX\_ASG[1] defines that 2 Holding Registers from address 1 to 2 are assigned to robot registers from R[1] multiplied by 1 as 16 bits signed integer. One Holding Register address is 16 bits data, and one register uses one Holding Register. So, address 1 is assigned to R[1] as 16 bits signed integer and address 2 is assigned to R[2] as 16bits signed integer.

\$SNPX\_ASG[2] defines that 4 Holding Registers from address 3 to 6 are assigned to robot registers from R[1] multiplied by 100 as 32 bits signed integer. One robot register uses 2 Holding Registers. So, addresses 3-4 are assigned to R[1] multiplied by 100 as 32 bits signed integer and address 5-6 are assigned to R[2] multiplied by 100 as 32 bits signed integer.

\$SNPX\_ASG[3] defines that 4 Holding Registers from 7 to 10 are assigned to robot registers from R[2] divided by 10 as 32 bits signed integer. One robot register uses 2 Holding Registers. So, addresses 7-8 are assigned to R[2] divided by 10 as 32 bits signed integer and address 9-10 are assigned to R[3] divided by 10 as 32 bits signed integer.

\$SNPX\_ASG[4] defines that 2 Holding Registers from 11 to 12 are assigned to robot register R[1] as 32bit REAL. So, addresses 11 to 12 are assigned to R[1] as 32bits REAL. OPC UA clients can read this bit array as Int16, so the OPC UA client should change this value to REAL number.

### 55.3.3 Assign Position Registers

\$SNPX\_ASG setting to assign Position Registers to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One Position Register uses 50 Holding Registers.<br>(You can change the number of Holding Registers for one position register by adding "@" in \$VAR_NAME)<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Please set "PR[1]" to assign Position Register PR[1]. The number in [ ] is the index of position register.<br><br>Continued position registers like PR[2]-PR[5] can be assigned by one \$SNPX_ASG element. In this case, please set \$SIZE 200 because the number of position registers to assign is 4. And please set \$VAR_NAME "PR[2]". The index 2 means the starting index of the continuous assignment.<br><br>In multi group system, PR[1] means group 1 data of PR[1]. Please set PR[G2:1] to assign group 2 data of PR[1].<br><br>If "@" is added just after the string, the specified part of the position register is assigned. This is explained later.                                                                                      |
| \$MULTIPLY             | Meaning: Multiplier<br>Only REAL values like X, Y, Z and J1 are effected by this setting. Refer to the table below to see which elements are affected.<br>The value accessed by an HMI device etc. is multiplied by \$MULTIPLY.<br>When \$MULTIPLY is 0, it means special that an HMI device etc. can access Holding Registers as 32bits REAL data.<br>When it is not 0, an HMI device etc. accesses Holding Registers as 32bits signed integer. And value is rounded off to no decimal place.<br>Range: 0.0001~10000, 0<br>Example: Position register member value is 123.45.<br>When \$MULTIPLY is 1, Holding Register is 123<br>When \$MULTIPLY is 10, Holding Register is 1235.<br>When \$MULTIPLY is 0.1, Holding Register is 12.<br>When \$MULTIPLY is 0, Holding Register is 123.45 of REAL |

**WARNING**

If Integer data is written to the Holding Registers of Real data type, the assigned robot data may be set to the value that is too big or cannot be processed. And it may cause serious effect such as the robot moves to unexpected position because the assigned position register has abnormal position data. Please set \$MULTIPLY to the value of not 0 normally, and assign 32 bit signed integer data to Holding Registers.

**NOTE**

When the position register is locked by LOCK PREG instruction, the writing to the position register by this function is ignored. To confirm if the writing is succeeded or not, please read the position data just after the writing and confirm the read data.

One position register uses 50 Holding Registers. Contents of the 50 Holding Registers are the following.

| Address        | Description |                                            | Effect of \$MULTIPLY |
|----------------|-------------|--------------------------------------------|----------------------|
| Cartesian data |             |                                            |                      |
| 1-2            | X           | 32bits signed integer or real (mm)         | Yes                  |
| 3-4            | Y           | 32bits signed integer or real (mm)         | Yes                  |
| 5-6            | Z           | 32bits signed integer or real (mm)         | Yes                  |
| 7-8            | W           | 32bits signed integer or real (deg)        | Yes                  |
| 9-10           | P           | 32bits signed integer or real (deg)        | Yes                  |
| 11-12          | R           | 32bits signed integer or real (deg)        | Yes                  |
| 13-14          | E1          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 15-16          | E2          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 17-18          | E3          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 19             | FLIP        | 16bits signed integer (1:Flip, 0:Non flip) | No                   |
| 20             | LEFT        | 16bits signed integer (1:Left, 0:Right)    | No                   |
| 21             | UP          | 16bits signed integer (1:Up, 0:Down)       | No                   |
| 22             | FRONT       | 16bits signed integer (1:Front, 0:Back)    | No                   |
| 23             | TURN4       | 16bits signed integer (-128~127)           | No                   |
| 24             | TURN5       | 16bits signed integer (-128~127)           | No                   |
| 25             | TURN6       | 16bits signed integer (-128~127)           | No                   |
| 26             | VALIDC      | 16bits signed integer (→NOTE1)             | No                   |
| Joint data     |             |                                            |                      |
| 27-28          | J1          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 29-30          | J2          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 31-32          | J3          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 33-34          | J4          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 35-36          | J5          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 37-38          | J6          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 39-40          | J7          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 41-42          | J8          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 43-44          | J9          | 32bits signed integer or real (mm, deg)    | Yes                  |
| 45             | VALIDJ      | 16bits signed integer (→NOTE2)             | No                   |
| Frame number   |             |                                            |                      |
| 46             | UF          | 16bits signed integer (-1~62) (→NOTE3)     | No                   |
| 47             | UT          | 16bits signed integer (-1~30) (→NOTE4)     | No                   |
| 48-50          | Reserve     |                                            | No                   |

**NOTE**

- 1 VALIDC shows whether the position data is valid for Cartesian or not. This becomes 0 at the following situation, and it becomes 1 in the other situation.
  - Position data has uninitialized data (displayed as "\*\*\*\*\*" on Teach Pendant).
  - Position data is Joint representation and it can not be converted to Cartesian.
 If HMI device etc. writes any data to VALIDC, position representation is changed to Cartesian. In this case, any value is OK to write.
- 2 VALIDJ shows whether the position data is valid for Joint or not. This becomes 0 at the following situation, and it becomes 1 in the other situation.
  - Position data has uninitialized data (displayed as "\*\*\*\*\*" on Teach Pendant).
  - Position data is Cartesian representation and it can not be converted to Joint.
 If HMI device etc. writes any data to VALIDJ, position representation is changed to Joint. In this case, any value is OK to write.
- 3 UF is user frame number.  
 If UF is 0, world frame is used.  
 If UF is -1, the current selected user fame is used.  
 UF of Position Register is always -1. (In iPendant it is displayed as F)  
 This value cannot be changed.
- 4 UT is tool frame number  
 If UT is 0, mechanical interface frame is used.  
 If UT is -1, the current selected tool fame is used.  
 UT of Position Register is always -1. (In iPendant it is displayed as F)  
 This value cannot be changed.

- Position register has 2 representations, Cartesian or Joint. If detail of the position register is displayed as X,Y,Z,W,P,R, it is Cartesian representation. If it is displayed as J1-6, it is Joint representation.
- An HMI device etc. can always access to any member without regarding to current representation. Position representation is changed by reading from HMI device etc. But if the representation of position register and the representation of accessed member from an HMI device etc. are different, please note the following.
- If the position data is out of stroke limit or there is uninitialized member, this position data cannot be converted to another representation. In this case, the all members in the part of another representation becomes 0.  
 For example, a position register is Cartesian representation and X is 10000, it is out of stroke limit, J1-J9 is read as 0 by an HMI device etc. In this case, the members of Cartesian part or UF/UT part can be read correctly.
- If the representation of position register and the representation of accessed member from an HMI device etc. are different, communication response time is increased. Because position representation conversion takes much time. If an HMI device etc. reads many position registers, this conversion time may make big effect to response time.

Uninitialized member of position data is displayed as "\*\*\*\*\*" on position register menu of Teach Pendant. These members are read as 0 from an HMI device etc. Please read VALIDC or VALIDJ to check whether the value is 0 or uninitialized. If position data has uninitialized member, VALIDJ and VALIDC are 0.

If you write data to the member in Cartesian part from an HMI device etc., the position representation becomes Cartesian. If you write data to member in Joint part from an HMI device etc., the position representation becomes Joint. If you write data to UF or UT from an HMI device etc., position representation is not changed.

**Extract a part by adding "@" in \$VAR\_NAME**

If "@" is added to \$VAR\_NAME, the specified part of data structure is extracted. It was already used for Register assignment (R[1]@1.1).



For example, you need to access only X, Y and Z of PR[1-3]. In this case, normally the setting of \$SNPX\_ASG is the following.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 150    | PR[1]      | 1          |

The following is the correspondence between position registers and Holding Registers of this setting.

| Address | Assigned data                       |
|---------|-------------------------------------|
| 1-2     | X of PR[1] as 32bits signed integer |
| 3-4     | Y of PR[1] as 32bits signed integer |
| 5-6     | Z of PR[1] as 32bits signed integer |
| 51-52   | X of PR[2] as 32bits signed integer |
| 53-54   | Y of PR[2] as 32bits signed integer |
| 55-56   | Z of PR[2] as 32bits signed integer |
| 101-102 | X of PR[3] as 32bits signed integer |
| 103-104 | Y of PR[3] as 32bits signed integer |
| 105-106 | Z of PR[3] as 32bits signed integer |

Actual read data are only 18 Holding Registers, but this assignment occupies 150 Holding Registers. And, some HMI devices access address 7-50 of Holding Registers that is not necessary to read, because reading one big data is more efficient than reading several small data from the communication point of view. But the address 27-45 includes Joint representation part, and if position data is Cartesian representation, representation conversion is needed to read this unnecessary data.

To communicate efficiently, please set \$SNPX\_ASG as follows.

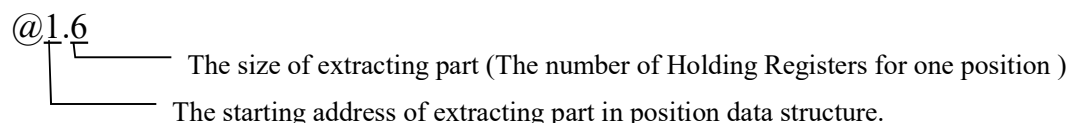
|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 18     | PR[1]@1.6  | 1          |

The following is the correspondence between position registers and Holding Registers of this setting.

| Address | Accessed data                       |
|---------|-------------------------------------|
| 1-2     | X of PR[1] as 32bits signed integer |
| 3-4     | Y of PR[1] as 32bits signed integer |
| 5-6     | Z of PR[1] as 32bits signed integer |
| 7-8     | X of PR[2] as 32bits signed integer |
| 9-10    | Y of PR[2] as 32bits signed integer |
| 11-12   | Z of PR[2] as 32bits signed integer |
| 13-14   | X of PR[3] as 32bits signed integer |
| 15-16   | Y of PR[3] as 32bits signed integer |
| 17-18   | Z of PR[3] as 32bits signed integer |

The change of \$SNPX\_ASG is that “@1.6” is added in \$VAR\_NAME and \$SIZE is changed from 150 to 18. By this change, the number of Holding Registers for one position register is changed from 50 to 6. The “6” in “@1.6” means the number of Holding Registers for one position register. And the “1” in “@1.6” means the starting address of extracting part in position data structure.

By specifying “@” in \$VAR\_NAME, you can extract the specified part of position data structure.



You can specify “@” not only for position register, but also for all data that is assigned by \$SNPX\_ASG.

The following is another example to assign J1-J6 of PR[1-3].

|               | \$ADDRESS | \$SIZE | \$VAR_NAME  | \$MULTIPLY |
|---------------|-----------|--------|-------------|------------|
| \$SNPX_ASG[1] | 1         | 96     | PR[3]@27.12 | 1          |

### 55.3.4 Assign String Registers

\$SNPX\_ASG setting to assign String Registers to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One String Register uses 40 Holding Registers.<br>It is possible to deal with first 80 characters of one string register in default setting.<br>(You can change the number of Holding Registers for one string register by adding "@" in \$VAR_NAME)<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Please set "SR[1]" to assign String Register SR[1]. The number in [ ] is the index of string register.<br><br>Continued string registers such as SR[2]-SR[5] can be assigned by one \$SNPX_ASG element. In this case, please set \$SIZE 160 because the number of string registers to assign is 4. And please set \$VAR_NAME "SR[2]". The index 2 means the starting index of the continuous assignment.<br><br>If "@" is added just after the string, the specified part of the String Register is assigned.<br>Example: SR[1]@1.50<br>"SR[1]" means to assign String Registers from index 1.<br>"@1.50" means the first 100 characters of String Register data are assigned. |
| \$MULTIPLY             | Meaning: Specify byte order of string.<br>Range: 1,-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### 55.3.5 Assign Current Position

\$SNPX\_ASG setting to assign Current Position to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                              |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>Current Position uses 50 Holding Registers.<br>(You can change the number of Holding Registers for Current Position by adding "@" in \$VAR_NAME)<br>Range: 1~16384 |

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$VAR_NAME             | <p>Meaning: String to specify robot data.<br/>Please set "POS[0]" to assign Current Position. The number in [ ] is user frame number.</p> <p>When user frame number is 0, the Current Position on World frame is assigned. This is the same as when "WORLD" is selected in the current position screen on the teaching pendant.</p> <p>When user frame number is -1, the Current Position on the User frame that is currently selected. This is the same as when "USER" is selected in the current position screen on the teaching pendant.</p> <p>When user frame number is 1-61, the Current Position on the specified user frame.</p> <p>Data structure is the same as position register.</p> <p>In multi group system, POS[0] means current position group 1 robot. Please set POS[G2:0] to assign current position of group 2 robot.</p> <p>If "@" is added just after the string, the specified part of current position is assigned.</p> |
| \$MULTIPLY             | <p>Meaning: Multiplier<br/>Only REAL values like X, Y, Z and J1 are effected by this setting. Refer to the table below to see which elements are affected.<br/>The value of each element of the current position multiplied by the value of \$MULTIPLY is read and written.</p> <p>If \$MULTIPLY is 0, it means that the data is assigned to Holding Registers as 32bits REAL data.<br/>If it is not 0, it means that the data is assigned to Holding Registers as 32bits signed integer. And value is rounded off to no decimal place.</p> <p>Range: 0.0001~10000, 0<br/>Example: Current position member value is 123.45.<br/>When \$MULTIPLY is 1, Holding Register is 123<br/>When \$MULTIPLY is 10, Holding Register is 1235.<br/>When \$MULTIPLY is 0.1, Holding Register is 12.<br/>When \$MULTIPLY is 0, Holding Register is 123.45 of REAL</p>                                                                                         |

**NOTE**

Current position is read only. If you write it, nothing occurs.

Current position uses 50 Holding Registers, and data structure is the same as position register.

UT of current position is always -1.

UF of current position is the specified user frame number in \$VAR\_NAME.

Current position is not assigned continuously. For example, the following setting of \$SNPX\_ASG assigns current position of user frame 1 to address 1-50, but current position of user frame 2 is not assigned to address 51-100. Please set "POS[2]" in another \$SNPX\_ASG.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 100    | POS[1]     | 1          |

### 55.3.6 Assign Alarm History

\$SNPX ASG setting to assign Alarm history to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One alarm uses 100 Holding Registers.<br>(You can change the number of Holding Registers for one alarm by adding "@" in \$VAR_NAME)<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Please set "ALM[1]" to assign alarm. The number in [ ] is line number in alarm menu. The latest alarm is 1.<br><br>"ALM[1]" assigns active alarm. Alarms displayed in active alarm menu are assigned.<br><br>"ALM[E1]" assigns alarm history. Alarms displayed in alarm history menu are assigned.<br><br>"ALM[M1]" assigns motion alarm history. Alarm displayed in motion alarm menu are assigned.<br><br>"ALM[S1]" assigns system alarm history. Alarm displayed in system alarm menu are assigned.<br><br>"ALM[A1]" assigns application alarm history. Alarm displayed in application alarm menu are assigned.<br><br>"ALM[P1]" assigns password log. Password log displayed in password log menu are assigned. |
| \$MULTIPLY             | Meaning: Specify byte order of string.<br>Range: 1,-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

#### NOTE

Alarm history is read only. If you write it, nothing occurs.

One alarm uses 100 Holding Registers. Contents of the 100 Holding Registers are the following.

| Address | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Facility code      16 bits signed integer<br>When alarm is "SRVO-001", facility code is 11 that means "SRVO". Please refer the "2.2.2 Facility Name and Code" in "OPERATOR'S MANUAL (Alarm Code List)" (B-83284EN-1) about the number of facility code.                                                                                                                                                                                                                                                                                            |
| 2       | Alarm number      16 bits signed integer<br>When alarm is "SRVO-001", alarm number is 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3       | Facility code of Cause Code      16bits signed integer<br>When alarm occurs, alarm message is displayed on top of teach pendant. Sometimes a message is also displayed on the second line, it is cause code.<br>This shows facility code of the cause code. If the alarm does not have cause code, this becomes 0.<br>For example, when cause code is "SRVO-001", facility code is 11 that means "SRVO". Please refer the "2.2.2 Facility Name and Code" in "OPERATOR'S MANUAL (Alarm Code List)" (B-83284EN-1) about the number of facility code. |
| 4       | Alarm number of cause code      16 bits signed integer<br>Alarm number of cause code. If the alarm does not have cause code, this becomes 0.<br>When cause code is "SRVO-001", alarm number of cause code is 1.                                                                                                                                                                                                                                                                                                                                    |

| Address | Description                                                                                                                                                                                                                                                                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5       | Alarm severity      16 bits signed integer<br>The value shows alarm severity.<br>NONE                128<br>WARN                0<br>PAUSE.L            2<br>PAUSE.G            34<br>STOP.L              6<br>STOP.G              38<br>SERVO               54<br>ABORT.L            11<br>ABORT.G            43<br>SERVO2             58<br>SYSTEM             123 |
| 6       | Occurred Time (year)      16 bits signed integer                                                                                                                                                                                                                                                                                                                     |
| 7       | Occurred Time (month)    16 bits signed integer                                                                                                                                                                                                                                                                                                                      |
| 8       | Occurred Time (day)      16 bits signed integer                                                                                                                                                                                                                                                                                                                      |
| 9       | Occurred Time (hour 24 hour system)    16 bits signed integer                                                                                                                                                                                                                                                                                                        |
| 10      | Occurred Time (minutes)   16 bits signed integer                                                                                                                                                                                                                                                                                                                     |
| 11      | Occurred Time (second)    16 bits signed integer                                                                                                                                                                                                                                                                                                                     |
| 12-51   | Alarm message      string of up to 80 characters<br>Alarm message string can be read. It shows the string as the same as that is displayed on top line of teach pendant such as "SRVO-001".                                                                                                                                                                          |
| 52-91   | Cause code alarm message    string of up to 80 characters<br>Alarm message of cause code.                                                                                                                                                                                                                                                                            |
| 92-100  | Alarm severity word    80 characters string<br>String of alarm severity such as "WARN".                                                                                                                                                                                                                                                                              |

Facility code and alarm number of "RESET" is read as 0. Alarm message is "RESET".

When there is only 2 lines on active alarm menu, all members of alarms after ALM[3] are 0.

In string item, addresses after string are 0.

One alarm uses 100 Holding Registers, if you would like to read only facility code and alarm number, most of 100 Holding Registers area is not used. To communicate efficiently, please use "@".

Example: \$SNPX ASG is set as follows.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME  | \$MULTIPLY |
|---------------|-----------|--------|-------------|------------|
| \$SNPX ASG[1] | 1         | 12     | ALM[E1]@1.4 | 1          |

The correspondence of address and alarm data of this setting is the following.

| Address | Assigned data                        |
|---------|--------------------------------------|
| 1       | Alarm ID of alarm1                   |
| 2       | Alarm number of alarm1               |
| 3       | Alarm ID of cause code of alarm1     |
| 4       | Alarm number of cause code of alarm1 |
| 5       | Alarm ID of alarm2                   |
| 6       | Alarm number of alarm2               |
| 7       | Alarm ID of cause code of alarm2     |
| 8       | Alarm number of cause code of alarm2 |
| 9       | Alarm ID of alarm3                   |
| 10      | Alarm number of alarm3               |
| 11      | Alarm ID of cause code of alarm3     |
| 12      | Alarm number of cause code of alarm3 |

### 55.3.7 Assign Program Execution Status

\$SNPX\_ASG setting to assign program execution status to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One task uses 18 Holding Registers.<br>(You can change the number of Holding Registers for one task by adding "@" in \$VAR_NAME)<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                      |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Please set "PRG[1]" to assign program execution status. The number in [ ] is task number.<br>In single task system, program execution status always can be assigned by PRG[1].<br><br>In multi task system, when 2 tasks are running at the same time, PRG[1] and PRG[2] shows the execution status of each task.<br>Which task is PRG[1] is decided by timing of execution and communication. But the task that is read as PRG[1] is always read as PRG[1] until it is aborted. |
| \$MULTIPLY             | Meaning: Specify byte order of string.<br>Range: 1,-1                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

#### NOTE

Program execution status is read only. If you write it, nothing occurs.

One task uses 18 Holding Registers. Contents of the 18 Holding Registers are the following.

| Address | Description                                                                                                                                                                                         |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1-8     | Program name 16 characters string (Long program name is truncated to 15 characters + NULL)<br>Name of running program.<br>When sub program is called, it shows sub program name.                    |
| 9       | Line number 16bits signed integer.<br>Execution line number.<br>When sub program is called, it shows line number of sub program.                                                                    |
| 10      | Execution status 16bits signed integer<br>Aborted 0<br>Paused 1<br>Running 2                                                                                                                        |
| 11-18   | Parent program name 16 characters string (Long program name is truncated to 15 characters + NULL)<br>Name of the started program<br>When sub program is not called, it is the same as program name. |

When program is aborted, all members become 0.

By assigning each strings to \$VAR\_NAME, the type of the program can be selected. And the meaning of line number is changed depending on the situation.

| \$VAR_NAME | Program name and line number                                                      |
|------------|-----------------------------------------------------------------------------------|
| PRG[1]     | Program name<br>Name of running program.<br>Line number<br>Execution line number. |

| <b>\$VAR_NAME</b>          | <b>Program name and line number</b>                                                                                                                                                                                                                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PRG[M1]                    | <p>Program name<br/>When running program is MACRO program, it shows name of parent program that calls the running program.<br/>When the parent program is also MACRO program, it shows name of parent program that is not MACRO program.</p> <p>Line number<br/>Line number of CALL instruction that calls the running program. When all parent programs are MACRO program. It becomes 0.</p>                                      |
| PRG[K1]                    | <p>Program name<br/>When running program is KAREL program, it shows name of parent program that calls the running program.<br/>When the parent program is also KAREL program, it shows name of parent program that is not KAREL program.</p> <p>Line number<br/>Line number of CALL instruction that calls the running program. When all parent programs are KAREL program. It becomes 0.</p>                                      |
| PRG[MK1]<br>or<br>PRG[KM1] | <p>Program name<br/>When running program is MACRO or KAREL program, it shows name of parent program that calls the running program.<br/>When the parent program is also MACRO or KAREL program, it shows name of parent program that is not MACRO nor KAREL program.</p> <p>Line number<br/>Line number of CALL instruction that calls the running program. When all parent programs are MACRO or KAREL program. It becomes 0.</p> |

By adding “@”, a part of this structure is assigned to Holding Registers.

Example: \$SNPX\_ASG is set as follows.

|               | <b>\$ADDRESS</b> | <b>\$SIZE</b> | <b>\$VAR_NAME</b> | <b>\$MULTIPLY</b> |
|---------------|------------------|---------------|-------------------|-------------------|
| \$SNPX_ASG[1] | 1                | 4             | PRG[1]@9.2        | 1                 |

The correspondence of address and program execution status of this setting is the following.

| <b>Address</b> | <b>Accessed data</b>        |
|----------------|-----------------------------|
| 1              | Line number of task 1       |
| 2              | Execution status of task 1  |
| 3              | Line number of task 2.      |
| 4              | Execution status of task 2. |

### 55.3.8 Assign System Variables

\$SNPX\_ASG setting to assign system variables to Holding Registers is the following.

| <b>Variable in \$SNPX_ASG</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                        |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS                     | <p>Meaning: Start address of Holding Registers to assign.<br/>Range: 1~16384</p>                                                                                                                                                                                                                                                          |
| \$SIZE                        | <p>Meaning: Number of Holding Registers to assign.<br/>The number of Holding Registers for one system variable is defined by data type of the system variable. Please refer the following data type list.<br/>(You can change the number of Holding Registers for one system variable by adding “@” in \$VAR_NAME)<br/>Range: 1~16384</p> |

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$VAR_NAME             | <p>Meaning: String to specify robot data.<br/>Please set system variable name, for example "\$SNPX_ASG[1].\$ADDRESS".</p> <p>If array system variable such as "\$SNPX_ASG[1]" is set, array elements are assigned continuously like Registers.</p> <p>KAREL string can be specified by the following format.<br/>\$[KAREL program name]variable name.</p> |
| \$MULTIPLY             | Meaning: The meaning of \$MULTIPLY is defined according to the data type of system variable. Please refer the following data type list.                                                                                                                                                                                                                   |

To access to system variable, data type of assigned Holding Registers and meaning of \$MULTIPLY are changed according to data type of system variable. The following is the list of data type of system variables that can be assigned to Holding Registers, and it also explain the number of Holding Registers and meaning of \$MULTIPLY.

| Data type of system variables    | The number of Holding Registers for one system variable | Meaning of \$MULTIPLY                                                                                                                                                                                                                                                                              |
|----------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| INTEGER<br>32bits signed integer | 2                                                       | <p>The data is multiplied by \$MULTIPLY.<br/>The data is accessed as 32 bits signed integer. And value is rounded off to no decimal place.<br/>If \$MULTIPLY is 0, it is same as \$MULTIPLY is 1.</p>                                                                                              |
| SHORT<br>16bits signed integer   | 2                                                       |                                                                                                                                                                                                                                                                                                    |
| BYTE<br>8bits signed integer     | 2                                                       |                                                                                                                                                                                                                                                                                                    |
| REAL<br>32bits real              | 2                                                       | <p>The data is multiplied by \$MULTIPLY.<br/>If \$MULTIPLY is 0, it means that the data is accessed as 32 bits REAL data.<br/>If it is not 0, the data is accessed as 32 bits signed integer. And value is rounded off to no decimal place</p>                                                     |
| BOOLEAN<br>TRUE/FALSE            | 2                                                       | <p>\$MULTIPLY is not used for this data type.<br/>The data is accessed as 32bit signed integer.<br/>If value is TRUE, Holding Register is 1. If it is 0, Holding Register is 0.<br/>If 0 is written, the system variable becomes FALSE. If not 0 is written, the system variable becomes TRUE.</p> |
| POSITION<br>Position data        | 50                                                      | <p>Data structure is the same as position register.<br/>Meaning of \$MULTIPLY is also same as position register.</p>                                                                                                                                                                               |
| STRING<br>String data            | 40                                                      | Specify byte order of string.                                                                                                                                                                                                                                                                      |

INTEGER, SHORT and BYTE are accessed by the same way. The system variable that integer value is displayed in system variable menu on teach pendant is INTEGER, SHORT or BYTE.

The variable that real value is displayed in system variable menu on teach pendant is REAL.

The variable that TRUE or FALSE is displayed in system variable menu on teach pendant is BOOLEAN.

The variable that string is displayed on system variable menu on teach pendant is STRING.

The variable that is displayed as POSITION or Recorded in system variable menu on teach pendant is POSITION. Structure other than POSITION can't assign with 1 assignment.



**⚠ WARNING**

The system variable that is protected in system variable menu on teach pendant can be changed from Holding Registers. If illegal value is set to system variable, system may be harmed seriously. Please check value enough to write system variables.

**⚠ WARNING**

If Integer data is written to the Holding Registers of Real data type, the assigned robot data may be set to the value that is too big or cannot be processed. And it may cause serious effect such as the robot moves to unexpected position because the assigned POSITION type variable has abnormal position data. Please set \$MULTIPLY to the value of not 0 normally, and assign 32 bit signed integer data to Holding Registers.

By adding “@”, a part of this structure is assigned to Holding Registers.

For example, the following \$SNPX ASG assigns X, Y and Z of user frame 1 and 2 of group 1.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME          | \$MULTIPLY |
|---------------|-----------|--------|---------------------|------------|
| \$SNPX ASG[1] | 1         | 12     | \$MNUFRAME[1,1]@1.6 | 1          |

This assigns system variables to Holding Registers as follows.

| Address | Accessed data                                       |
|---------|-----------------------------------------------------|
| 1-2     | X of user frame 1 of Group 1 (X of \$MNUFRAME[1,1]) |
| 3-4     | Y of user frame 1 of Group 1 (Y of \$MNUFRAME[1,1]) |
| 5-6     | Z of user frame 1 of Group 1 (Z of \$MNUFRAME[1,1]) |
| 7-8     | X of user frame 2 of Group 1 (X of \$MNUFRAME[1,2]) |
| 9-10    | Y of user frame 2 of Group 1 (Y of \$MNUFRAME[1,2]) |
| 11-12   | Z of user frame 2 of Group 1 (Z of \$MNUFRAME[1,2]) |

## Special system variables

The following table shows the list of the special system variables that are not actual system variables but it can be assigned to Holding Registers to read the robot internal information. The special system variables are read only. The value is not changed, when the assigned Holding Registers is written. The special system variables of \$RBTIF.MEM\_~ are supported from 7DC1 series P06 or later. The other special system variables are supported from 7DC3 series P43 or later and 7DF1 series P15 or later and all versions after 7DF1 series.

| Variable name         | Data Type | Description                                                                                                     |
|-----------------------|-----------|-----------------------------------------------------------------------------------------------------------------|
| \$RBTIF.MEM_TPP_TOT   | INTEGER   | Meaning: Total storage capacity of the TPP.<br>The unit displayed in the holding register is byte.              |
| \$RBTIF.MEM_TPP_FREE  | INTEGER   | Meaning: Available storage capacity of the TPP.<br>The unit displayed in the holding register is byte.          |
| \$RBTIF.MEM_TEMP_TOT  | INTEGER   | Meaning: Total storage capacity of the TEMP.<br>The unit displayed in the holding register is byte.             |
| \$RBTIF.MEM_TEMP_FREE | INTEGER   | Meaning: Available storage capacity of the TEMP.<br>The unit displayed in the holding register is byte.         |
| \$RBTIF.MEM_PERM_TOT  | INTEGER   | Meaning: Total storage capacity of the PERM.<br>The unit displayed in the holding register is byte.             |
| \$RBTIF.MEM_PERM_FREE | INTEGER   | Meaning: Available storage capacity of the PERM.<br>The unit displayed in the holding register is byte.         |
| \$RBTIF.MEM_FROM_TOT  | INTEGER   | Meaning: Total storage capacity of the F-ROM module.<br>The unit displayed in the holding register is byte.     |
| \$RBTIF.MEM_FROM_FREE | INTEGER   | Meaning: Available storage capacity of the F-ROM module.<br>The unit displayed in the holding register is byte. |

| Variable name                                      | Data Type | Description                                                                                                                                                                                                 |
|----------------------------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$RBTIF.SERIAL_NUM                                 | STRING    | Meaning: Robot serial number.                                                                                                                                                                               |
| \$RBTIF.SOFT_VERSION                               | STRING    | Meaning: Robot software version.                                                                                                                                                                            |
| \$RBTIF.ROBOT_ID_GRP1 ~<br>\$RBTIF.ROBOT_ID_GRP8   | STRING    | Meaning: Robot ID.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set.                                                                                |
| \$RBTIF.OVERRIDE                                   | INTEGER   | Meaning: Override.<br>Range is 1 to 100%.                                                                                                                                                                   |
| \$RBTIF.STIME_PWR_TOT1 ~<br>\$RBTIF.STIME_PWR_TOT8 | INTEGER   | Meaning: Total time of controller power on.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is minute. |
| \$RBTIF.STIME_SRV_TOT1 ~<br>\$RBTIF.STIME_SRV_TOT8 | INTEGER   | Meaning: Total time of servo on.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is minute.            |
| \$RBTIF.STIME_RUN_TOT1 ~<br>\$RBTIF.STIME_RUN_TOT8 | INTEGER   | Meaning: Total time of program running.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is minute.     |
| \$RBTIF.STIME_WIT_TOT1 ~<br>\$RBTIF.STIME_WIT_TOT8 | INTEGER   | Meaning: Total time of program waiting.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is minute.     |
| \$RBTIF.STIME_SHM_TOT1 ~<br>\$RBTIF.STIME_SHM_TOT8 | INTEGER   | Meaning: Total time of brake release.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is minute.       |
| \$RBTIF.STIME_PWR_LAP1 ~<br>\$RBTIF.STIME_PWR_LAP8 | INTEGER   | Meaning: Lap time of controller power on.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is second.   |
| \$RBTIF.STIME_SRV_LAP1 ~<br>\$RBTIF.STIME_SRV_LAP8 | INTEGER   | Meaning: Lap time of servo on.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is second.              |
| \$RBTIF.STIME_RUN_LAP1 ~<br>\$RBTIF.STIME_RUN_LAP8 | INTEGER   | Meaning: Lap time of program running.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is second.       |
| \$RBTIF.STIME_WIT_LAP1 ~<br>\$RBTIF.STIME_WIT_LAP8 | INTEGER   | Meaning: Lap time of program waiting.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is second.       |
| \$RBTIF.STIME_SHM_LAP1 ~<br>\$RBTIF.STIME_SHM_LAP8 | INTEGER   | Meaning: Lap time of brake release.<br>The last number corresponds to the number of groups.<br>If invalid last number is set, value 0 is set. The unit displayed in the holding register is second.         |

| Variable name                                                                                                                                         | Data Type | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$RBTIF.Q_CURRENT11 ~<br>\$RBTIF.Q_CURRENT19,<br>\$RBTIF.Q_CURRENT21 ~<br>\$RBTIF.Q_CURRENT29,<br>...<br>\$RBTIF.Q_CURRENT81 ~<br>\$RBTIF.Q_CURRENT89 | REAL      | Meaning: Q phase current of each axis. Since the torque constant is not disclosed, it cannot be converted to torque, but the Q-phase current is data proportional to the torque.<br><br>The last number corresponds to the number of groups and axis number. If invalid last number is set, value 0 is set. The unit displayed in the holding register is ampere.<br>Example:<br>\$SNPX_ASG.\$VAR_NAME : \$RBTIF.Q_CURRENT23<br>Group number is 2, Axis number is 3. |
| \$RBTIF.CURRENT11 ~<br>\$RBTIF.CURRENT19,<br>\$RBTIF.CURRENT21 ~<br>\$RBTIF.CURRENT29,<br>...<br>\$RBTIF.CURRENT81 ~<br>\$RBTIF.CURRENT89             | REAL      | Meaning: Current of each axis.<br><br>The last number corresponds to the number of groups and axis number. If invalid last number is set, value 0 is set. The unit displayed in the holding register is ampere.<br>Example:<br>\$SNPX_ASG.\$VAR_NAME : \$RBTIF.CURRENT23<br>Group number is 2, Axis number is 3.                                                                                                                                                     |

### 55.3.9 Assign comment

\$SNPX\_ASG setting to assign comment of Register, Position register, String register and I/O to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1 to 16384                                                                                                                                                                                              |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One comment uses 40 Holding Registers. Please set according to the number of comments to assign.<br>(You can change the number of Holding Registers for one comment by adding "@" in \$VAR_NAME)<br>Range: 1 to 16384 |

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------------------|--------|-----------------|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|----|--------|-----|---------|-----|---------|----|--------|----|--------|----|--------|----|--------|
| \$VAR_NAME             | <p>Meaning: String to specify robot data.</p> <p>Please set "R[C1]" to assign comment of register. The number in [ ] is index of register.</p> <p>To assign comment of position register, string register and I/O, please set the following string.</p> <table> <tr><td>Register</td><td>R[C1]</td></tr> <tr><td>Position register</td><td>PR[C1]</td></tr> <tr><td>String register</td><td>SR[C1]</td></tr> <tr><td>DI</td><td>DI[C1]</td></tr> <tr><td>DO</td><td>DO[C1]</td></tr> <tr><td>RI</td><td>RI[C1]</td></tr> <tr><td>RO</td><td>RO[C1]</td></tr> <tr><td>UI</td><td>UI[C1]</td></tr> <tr><td>UO</td><td>UO[C1]</td></tr> <tr><td>SI</td><td>SI[C1]</td></tr> <tr><td>SO</td><td>SO[C1]</td></tr> <tr><td>WI</td><td>WI[C1]</td></tr> <tr><td>WO</td><td>WO[C1]</td></tr> <tr><td>WSI</td><td>WSI[C1]</td></tr> <tr><td>WSO</td><td>WSO[C1]</td></tr> <tr><td>GI</td><td>GI[C1]</td></tr> <tr><td>GO</td><td>GO[C1]</td></tr> <tr><td>AI</td><td>AI[C1]</td></tr> <tr><td>AO</td><td>AO[C1]</td></tr> </table> <p>Comment of the continued registers such as R[2]~R[5] can be assigned by one \$SNPX_ASG element. In this case, please set \$SIZE 160 because the number for one comment to assign is 40. And please set \$VAR_NAME "R[C2]". The index 2 means the starting index of the continuous assignment.</p> <p>If "@" is added just after the string, the specified part of the comment is assigned.<br/> Example: R[C1]@1.5<br/> "R[C1]" means to assign comment of registers from index 1.<br/> "@1.5" means the first 10 characters of comment are assigned.</p> | Register | R[C1] | Position register | PR[C1] | String register | SR[C1] | DI | DI[C1] | DO | DO[C1] | RI | RI[C1] | RO | RO[C1] | UI | UI[C1] | UO | UO[C1] | SI | SI[C1] | SO | SO[C1] | WI | WI[C1] | WO | WO[C1] | WSI | WSI[C1] | WSO | WSO[C1] | GI | GI[C1] | GO | GO[C1] | AI | AI[C1] | AO | AO[C1] |
| Register               | R[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| Position register      | PR[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| String register        | SR[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| DI                     | DI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| DO                     | DO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| RI                     | RI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| RO                     | RO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| UI                     | UI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| UO                     | UO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| SI                     | SI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| SO                     | SO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| WI                     | WI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| WO                     | WO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| WSI                    | WSI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| WSO                    | WSO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| GI                     | GI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| GO                     | GO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| AI                     | AI[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| AO                     | AO[C1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |
| \$MULTIPLY             | <p>Meaning: Specify byte order of string.</p> <p>Range: 1,-1</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |       |                   |        |                 |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |    |        |     |         |     |         |    |        |    |        |    |        |    |        |

### 55.3.10 Assign I/O data and simulation status

\$SNPX\_ASG setting to assign I/O data and simulation status to Holding Registers is the following.  
 Input signal can't be changed from this function even though it is assigned to Holding Registers.  
 If you want to change input signal from this function, there is a way to change input signal through flag with assigning the input signal to flag, which is rack 34 and slot 1.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|------------|----|-------|--------|----|-------|--------|----|-------|--------|----|-------|--------|----|-------|--|----|-------|--|----|-------|--|----|-------|--|----|-------|--------|----|-------|--------|-----|--------|---------|-----|--------|---------|----|-------|--------|----|-------|--------|----|-------|--------|----|-------|--------|------|------|--|--------|------|--|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>The number of Holding Registers for one I/O value or simulation status is changed by \$MULTIPLY setting.<br>When \$MULTIPLY is 1, one I/O value or simulation status uses one Holding Register.<br>When \$MULTIPLY is 0, one Holding Register includes 16 I/O value or simulation status as bit image.<br>(Note: Value of GI/O and AI/O use one Holding Register even if \$MULTIPLY is 0.)<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Please set "DI[1]" to assign DI[1] data. The number in [ ] is index of DI.<br>To assign I/O data or simulation status of the other type of I/O, please set the following string. <table><tr><td></td><td>Value</td><td>Simulation</td></tr><tr><td>DI</td><td>DI[1]</td><td>DI[S1]</td></tr><tr><td>DO</td><td>DO[1]</td><td>DO[S1]</td></tr><tr><td>RI</td><td>RI[1]</td><td>RI[S1]</td></tr><tr><td>RO</td><td>RO[1]</td><td>RO[S1]</td></tr><tr><td>UI</td><td>UI[1]</td><td></td></tr><tr><td>UO</td><td>UO[1]</td><td></td></tr><tr><td>SI</td><td>SI[1]</td><td></td></tr><tr><td>SO</td><td>SO[1]</td><td></td></tr><tr><td>WI</td><td>WI[1]</td><td>WI[S1]</td></tr><tr><td>WO</td><td>WO[1]</td><td>WO[S1]</td></tr><tr><td>WSI</td><td>WSI[1]</td><td>WSI[S1]</td></tr><tr><td>WSO</td><td>WSO[1]</td><td>WSO[S1]</td></tr><tr><td>GI</td><td>GI[1]</td><td>GI[S1]</td></tr><tr><td>GO</td><td>GO[1]</td><td>GO[S1]</td></tr><tr><td>AI</td><td>AI[1]</td><td>AI[S1]</td></tr><tr><td>AO</td><td>AO[1]</td><td>AO[S1]</td></tr><tr><td>Flag</td><td>F[1]</td><td></td></tr><tr><td>Marker</td><td>M[1]</td><td></td></tr></table><br>Example: Assign DI[11]~DI[42]:<br>When \$MULTIPLY is 1, please set 32 to \$SIZE because the number of DI is 32, and set "DI[11]" to \$VAR_NAME. In this case, index 11 means that the start index of the assignment is 11.<br>When \$MULTIPLY is 0, 16 DI data are assigned to one Holding Register, so please set 2 to \$SIZE, and set "DI[11]" to \$VAR_NAME. |            | Value | Simulation | DI | DI[1] | DI[S1] | DO | DO[1] | DO[S1] | RI | RI[1] | RI[S1] | RO | RO[1] | RO[S1] | UI | UI[1] |  | UO | UO[1] |  | SI | SI[1] |  | SO | SO[1] |  | WI | WI[1] | WI[S1] | WO | WO[1] | WO[S1] | WSI | WSI[1] | WSI[S1] | WSO | WSO[1] | WSO[S1] | GI | GI[1] | GI[S1] | GO | GO[1] | GO[S1] | AI | AI[1] | AI[S1] | AO | AO[1] | AO[S1] | Flag | F[1] |  | Marker | M[1] |  |
|                        | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Simulation |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| DI                     | DI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | DI[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| DO                     | DO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | DO[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| RI                     | RI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RI[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| RO                     | RO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RO[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| UI                     | UI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| UO                     | UO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| SI                     | SI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| SO                     | SO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| WI                     | WI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | WI[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| WO                     | WO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | WO[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| WSI                    | WSI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | WSI[S1]    |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| WSO                    | WSO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | WSO[S1]    |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| GI                     | GI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GI[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| GO                     | GO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | GO[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| AI                     | AI[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AI[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| AO                     | AO[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AO[S1]     |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| Flag                   | F[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| Marker                 | M[1]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |
| \$MULTIPLY             | Meaning: One Holding Register has one I/O data or 16 I/O data as bit image.<br>One I/O data is assigned to one Holding Register 1<br>16 I/O data are assigned to one Holding Register 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |       |            |    |       |        |    |       |        |    |       |        |    |       |        |    |       |  |    |       |  |    |       |  |    |       |  |    |       |        |    |       |        |     |        |         |     |        |         |    |       |        |    |       |        |    |       |        |    |       |        |      |      |  |        |      |  |

You can select by \$MULTIPLY setting whether one I/O data is assigned to one Holding Register or 16 I/O data are assigned to one Holding Register.

When \$MULTIPLY is 1, one I/O data or one simulation status are assigned to one Holding Register. When the value is ON, the corresponded Holding Register is set to 1. When the value is OFF, the corresponded Holding Register is set to 0.

When \$MULTIPLY is 0, 16 I/O data or 16 simulation status are assigned to one Holding Register. When the value is ON, the corresponded bit of Holding Register is set to 1. When the value is OFF, the corresponded bit of Holding Register is set to 0. I/O of lower index is assigned to lower bit of Holding Register.

Example: When DI[1-16] is assigned to Holding Register, if only DI[1] is ON and the others are OFF, the Holding Register is 1. If only DI[16] is ON and the others are OFF, the Holding Register is 32768 (in case of unsigned 16 bit integer).

### 55.3.11 Assign Integrated PMC address data

\$SNPX\_ASG setting to Integrated PMC address data to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>The number of Holding Registers corresponded to PMC address is different by the specified PMC address that is byte address or bit address.<br>When bit address is specified, one bit PMC address is assigned to one Holding Register.<br>When byte address is specified, two bytes PMC addresses are assigned to one Holding Register.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                   |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Set "PMC1:X0" to assign data of PMC address X0. The "PMC1" part is PMC path number. "PMCS" means DCS Safety PMC.<br>Data of bit address is assigned by specifying bit address such as "PMC1:X0.0".<br>Example: Assign X0.0~X1.7 data of 1st path PMC<br>When \$SIZE is set to 1 and \$VAR_NAME is set to "PMC1:X0", two byte address X0~X1 are assigned to one Holding Register address. Lower byte is assigned to X0, upper byte is assigned to X1.<br>When \$SIZE is set to 16 and \$VAR_NAME is set to "PMC1:X0.0", 16 bit address X0.0~X1.7 are assigned to 16 Holding Registers. X0.0 is assigned to Address 1, X0.1 is assigned to Address 2, X1.7 is assigned to Address 16. |
| \$MULTIPLY             | Not used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

#### NOTE

- 1 When K900.4 (Change address value) is 0, writing to PMC address data is ignored. To check if the writing is ignored or not, please read the PMC address after writing and confirm the read value.
- 2 Address of DCS safety PMC is read only. If you write it, nothing occurs.

Example: \$SNPX\_ASG is set as follows.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 2      | PMC1:R0.0  | 1          |
| \$SNPX_ASG[2] | 3         | 2      | PMC1:D0    | 1          |

The correspondence between Holding Registers and robot data by this setting is the following.

Bit address is assigned to the address 1 - 2, therefore 1 bit of PMC address is assigned to 1 Holding Register continuously.

Byte address is assigned to the address 3 - 4, therefore 2 bytes of PMC address is assigned to 1 Holding Register continuously.

| Address | Assigned robot data                    |
|---------|----------------------------------------|
| 1       | R0.0                                   |
| 2       | R0.1                                   |
| 3       | D0 - D1 (Lower byte:D0, Upper byte:D1) |
| 4       | D2 - D3 (Lower byte:D2, Upper byte:D3) |

### 55.3.12 Assign Symbol and Comment of Integrated PMC address

\$SNPX\_ASG setting to symbol and comment of Integrated PMC address to Holding Registers is the following.

| Variable in \$SNPX_ASG | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| \$ADDRESS              | Meaning: Start address of Holding Registers to assign.<br>Range: 1~16384                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| \$SIZE                 | Meaning: Number of Holding Registers to assign.<br>One symbol or comment uses 40 Holding Registers. Please set according to the number of comments to assign.<br>(You can change the number of Holding Registers for one symbol and comment by adding "@" in \$VAR_NAME.)<br>Range: 1~16384                                                                                                                                                                                                                       |
| \$VAR_NAME             | Meaning: String to specify robot data.<br>Set "PMC1:S:X0" to assign symbol of PMC address X0. Set "PMC1:C:X0" to assign comment of the PMC address. The "PMC1" part is PMC path number. "PMCS" means DCS Safety PMC.<br>Symbol and comment of bit address is assigned by specifying bit address such as "PMC1:S:X0.0".<br>Example: PMC1:S:X0.0@1.5<br>The "PMC1:S:X0.0" part means that symbol of bit address from X0.0 is assigned continuously.<br>The @1.5 part means that the top 10 characters are assigned. |
| \$MULTIPLY             | Meaning: Specify byte order of string.<br>Range: 1,-1                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

#### NOTE

Symbol and comment of Integrated PMC are read only. If you write it, nothing occurs.

Example: \$SNPX\_ASG is set as follows.

|               | \$ADDRESS | \$SIZE | \$VAR_NAME      | \$MULTIPLY |
|---------------|-----------|--------|-----------------|------------|
| \$SNPX_ASG[1] | 1         | 10     | PMC1:S:R0.0@1.5 | 1          |
| \$SNPX_ASG[2] | 11        | 10     | PMC1:C:D0@1.5   | 1          |

The correspondence between Holding Registers and robot data by this setting is the following.

| Address | Assigned robot data |
|---------|---------------------|
| 1-5     | Symbol of R0.0      |
| 6-10    | Symbol of R0.1      |
| 11-15   | Comment of D0       |
| 16-20   | Comment of D1       |

### 55.3.13 Hints

#### Assigned data is not read correctly

If an HMI device etc. reads a Holding Register that is not assigned to any data, this Holding Register is always read as 0. If Holding Register that the problem occurs is not 0, this Holding Register is assigned to the other data.

For example, when \$SNPX\_ASG is set as follows, addresses 101-150 of Holding Registers are assigned by both \$SNPX\_ASG[1] and \$SNPX\_ASG[2].

|               | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|---------------|-----------|--------|------------|------------|
| \$SNPX_ASG[1] | 1         | 1000   | R[1]@1.1   | 1          |
| \$SNPX_ASG[2] | 101       | 50     | PR[1]      | 100        |

In this case, the \$SNPX\_ASG whose index is smaller is used.

Therefore, addresses 101-150 are assigned to R[101]-R[150], and PR[1] can not be accessed by an HMI device etc.

If the read data of Holding Register is 0, there may also be duplicated assignment as above. Please check duplication of \$SNPX\_ASG.

If \$SNPX\_ASG setting has problem even though assignment is not duplicated, please check \$VAR\_NAME of \$SNPX\_ASG.

The following list shows the correct format of \$VAR\_NAME setting. If the setting is not match to them, robot data is not assigned.

| Correct format                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Note                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| "R[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                               |
| "PR[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | If you specify group number, ":" must be specified between group number and index.                            |
| "PR[Gn:n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               |
| "SR[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                               |
| "POS[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | If you specify group number, ":" must be specified between group number and index.                            |
| "POS[Gn:n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                               |
| "ALM[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Alarm line number must be specified just after E, M, A, S or P. The ":" must not be specified.                |
| "ALM[En]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                               |
| "ALM[Pn]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                               |
| "PRG[n]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Index must be specified just after M, K, MK or KM. The ":" must not be specified.                             |
| "PRG[Mn]", "PRG[Kn]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                               |
| "PRG[MKn]", "PRG[KMn]"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                               |
| System variable name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | The first character of system variable must be "\$".                                                          |
| \$(KAREL program name)variable name                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Specify KAREL program name just after '\$[', then specify ']' and specify KAREL variable name just after ']'. |
| "DI[n]", "DI[Sn]", "DI[Cn]"<br>"DO[n]", "DO[Sn]", "DO[Cn]"<br>"RI[n]", "RI[Sn]", "RI[Cn]"<br>"RO[n]", "RO[Sn]", "RO[Cn]"<br>"UI[n]", "UI[Cn]"<br>"UO[n]", "UO[Cn]"<br>"SI[n]", "SI[Cn]"<br>"SO[n]", "SO[Cn]"<br>"WI[n]", "WI[Sn]", "WI[Cn]"<br>"WO[n]", "WO[Sn]", "WO[Cn]"<br>"WSI[n]", "WSI[Sn]", "WSI[Cn]"<br>"WSO[n]", "WSO[Sn]", "WSO[Cn]"<br>"GI[n]", "GI[Sn]", "GI[Cn]"<br>"GO[n]", "GO[Sn]", "GO[Cn]"<br>"AI[n]", "AI[Sn]", "AI[Cn]"<br>"AO[n]", "AO[Sn]", "AO[Cn]"<br>"F[n]", "F[Cn]"<br>"M[n]", "M[Cn]" | Index must be specified just after S or C. The ":" must not be specified.                                     |



| Correct format                                                                                                                                 | Note                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PMC<path>:address<br>PMC<path>:address.bit<br>PMC<path>:S:address<br>PMC<path>:S:address.bit<br>PMC<path>:C:address<br>PMC<path>:C:address.bit | The <path> is from 1 to 5 or "S". The ":" must be specified just after PMC path. And PMC address must be specified just after the ":". For symbol or comment, ":S" or ":C" must be specified just after PMC path. |

- If there is space character in \$VAR\_NAME, data is not assigned. If you specify "@", please do not add any space before "@".
- If you use continuous array assignment, the number of Holding Registers for one element is defined according to data type, please check explanation of the assigned robot data.
- The format of "@" is "@n.n". A "." must be specified between starting address and the number of Holding Registers. A "@" must be specified just after variable name.

### Method to improve response time

- Position representation conversion takes much time. If you access position register, please assign only necessary members of position structure by using "@". And please change position representation of the position registers to the representation that is accessed by an HMI device etc.

## 55.4 DYNAMIC ASSIGNMENT TO HOLDING REGISTERS

Setting \$SNPX\_ASG is necessary to read or write except for I/O signal. This setting is done by the system variable screen normally. However, dynamic settings, which is equivalent to \$SNPX\_ASG, is available for each connection after communication connection from Modbus master or from OPC UA client.

Please note that this setting is cleared when the connection is closed.

Settings are enabled only in the same connection so read and write Holding Registers with keeping connection or set every connection. Many masters / clients disconnect after every writing, if dynamic assignment doesn't work correctly, you have to understand and control master / client operation for the level of Modbus or TCP/IP protocol.

This setting takes precedence over \$SNPX\_ASG setting.

This function is supported from 7DF1 series P19 or after and all versions after 7DF1 series.

When 128 characters or less are written to the address 16385 or after of Holding Registers (It is necessary to start writing from address 16385) with Modbus from master or Command node with OPC UA from client, robot regards the written characters as command and processes it. Byte order of string data when Modbus can be changed with \$SNPX\_PARAM.\$CMD\_ENDIAN. (This doesn't affect to the byte order of Holding registers.)

Other than 0 (default) : Lower byte is the first character, upper byte is the second character.

0 : Upper byte is the first character, lower byte is the second character.

The following commands can be executed.

Each command should be written in 1 time. Write some times with segment is invalid.

### CLRASG Clear all assignment and initialization

Format: CLRASG

Function: Clear all dynamic assignment in the connection and initialization (prepare for using dynamic assignment).

Do once before following SETASG command sequence.

### SETASG Set assignment

Format: SETASG (\$ADDRESS) (\$SIZE) (\$VAR\_NAME) [ (\$MULTIPLY) ]

Function: Set assignment.

The specification of \$ADDRESS, \$SIZE, \$VAR\_NAME and \$MULTIPLY are same with the specification of the same variables under \$SNPX\_ASG.

\$MULTIPLY can be abbreviated. If abbreviated, \$MULTIPLY is set to 1.

Do CLRASG before executing SETASG sequence.

Example: If assignments are set as following

| No. | \$ADDRESS | \$SIZE | \$VAR_NAME | \$MULTIPLY |
|-----|-----------|--------|------------|------------|
| 1   | 1         | 2      | R[1]@1.1   | 1          |
| 2   | 3         | 4      | R[1]       | 100        |
| 3   | 7         | 4      | R[2]       | 0.1        |
| 4   | 11        | 2      | R[1]       | 0          |

Write following strings in order.

|                       |
|-----------------------|
| CLRASG                |
| SETASG 1 2 R[1]@1.1 1 |
| SETASG 3 4 R[1] 100   |
| SETASG 7 4 R[2] 0.1   |
| SETASG 11 2 R[1] 0    |

If some commands are separated with CR (13) or LF (10), some commands can be written in 1 sequence but the number of max character in 1 sequence is 128.

- There is no way to know what the current dynamic allocation is.
- Dynamic assignment allows assignment as long as memory allows.
- Even if allocation is not possible, no error is returned. If an address that cannot be assigned is read, 0 is read, and writing to it is ignored.

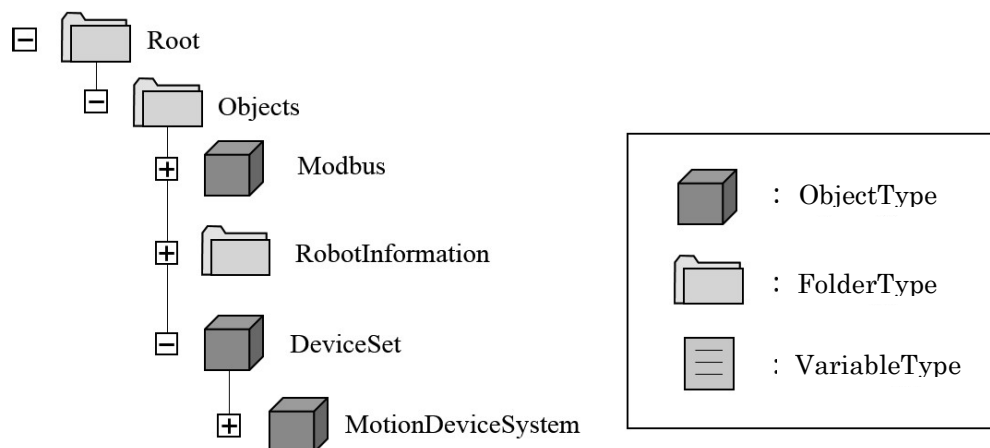
## 55.5 OPC UA DATA MODEL

OPC UA has ObjectType, FolderType and VariableType nodes.

ObjectType node is the pre-defined object, on the other hand, FolderType node just collects some nodes.

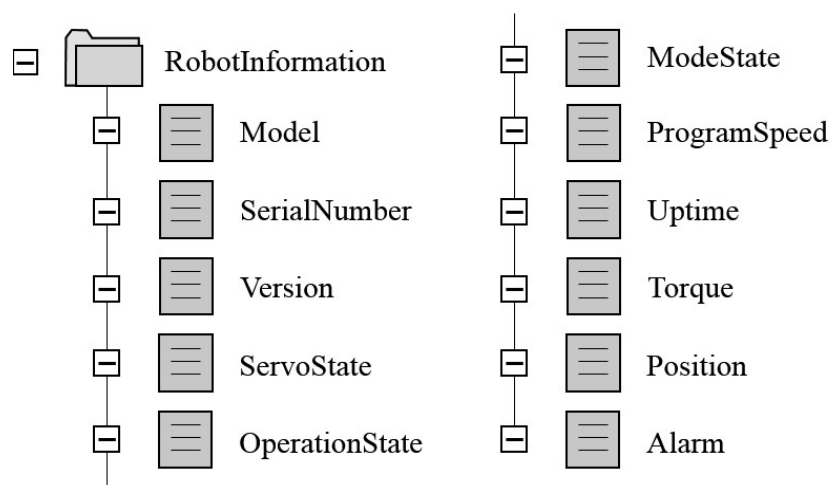
The OPC UA server for FANUC robots has Modbus node that provides read/write of robot information using the Modbus data model (see 55.2 MODBUS DATA MODEL and 55.3 ASSIGNMENT OF HOLDING REGISTERS), RobotInformation node that provides basic robot information, and MotionDeviceSystem node of the OPC UA for Robotics specification.

- RobotInformation node and MotionDeviceSystem node are supported from 7DF1 series P28 or after, 7DF3 series P06 or after, and all versions after 7DF3 series.



### 55.5.1 RobotInformation

The RobotInformation node provides basic information such as the robot model name and the robot controller software version. It has the following nodes.



#### Model String type

The Model node provides the model name for the group 1 robot.  
e.g. R-2000iC/165F

#### SerialNumber String type

The SerialNumber node provides the robot controller serial number.

#### Version String type

The Version node provides the software version of the robot controller.  
e.g. 7DF1/19

#### ServoState Byte type (1 byte)

The ServoState node provides the status of servo on / off.  
The node value is set to 1 when the servo is on and 0 when the servo is off.

**OperationState      Uint16 type (2 byte)**

The OperationState node provides status of E-STOP, Abort, PROGRUN, PAUSED, HELD, FAULT, TPENBL, BATALM and BUSY.

The value of each state is set in bit units, and the value converted to decimal is represented by one node value.

E-STOP is set to 1 when any one of the SOP E-STOP, TP E-STOP, and Ext E-STOP is input, and is set to 0 otherwise.

Abort is set to 1 when PROGRUN of the peripheral I / O (UO) is OFF and PAUSED is OFF.

PROGRUN ~ BUSY is the same as UO of peripheral I / O.

The OperationState node is set in the following order from the lower bits.

E-STOP (1) | Abort (2) | PROGRUN (4) | PAUSED (8) | HELD (16) | FAULT (32) | TPENBL (64) | BATALM (128) | BUSY (256)

**ModeState      String type**

The ModeState node provides the operation mode of the robot.

The current operation mode is set to the node value.

Operation modes are Auto, T1 (Manual), and T2 (Manual).

**ProgramSpeed      Byte type (1 byte)**

The ProgramSpeed node provides the speed override of robot.

The value from 0 to 100 (%) is set to the node value.

**Uptime      Int32 type (4 byte)**

The Uptime node provides the total energization time of the robot controller.

The value of the total energization time (min) is set to the node value.

**Torque      Float type (4 byte)**

The Torque node provides the Q-phase current value of each axis for the robot (until 6 axis) of group 1 and independent axis for group 1.

The Q-phase current value (A) of each axis is set to the node value.

**Position      Float type (4 byte)**

The Position node provides the position of each axis for the robot (until 6 axis) of group 1 and independent axis for group 1.

**Alarm      String type**

The Alarm node provides the current alarm.

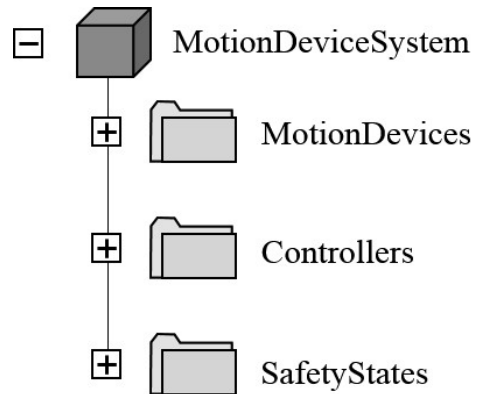
The current alarm string is set to the node value.

e.g. SRVO-001 Operator panel E-stop

**55.5.2      MotionDeviceSystem**

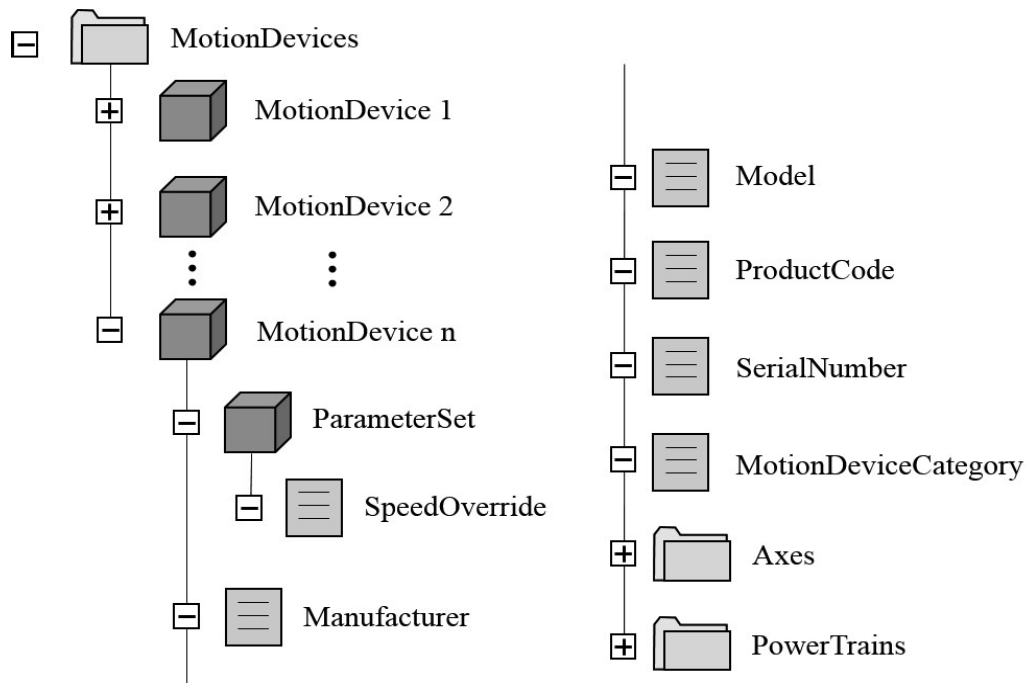
The MotionDeviceSystem node is the node based on OPC UA for Robotics.

It has nodes for MotionDevices, Controllers, and SafetyStatus.



### 55.5.2.1 MotionDevices

The MotionDevices node provides information such as robot model, overrides, axes, and motors. It has as many MotionDevice nodes as the number of robot groups.



#### **SpeedOverride** **Double type (8 byte)**

The SpeedOverride node provides the robot speed override.  
The value from 1 to 100 (%) is set to the node value.

#### **Manufacturer** **LocalizedText type**

The Manufacturer node provides the name of the robot manufacturer.  
"FANUC" is set to the node value.

#### **Model** **LocalizedText type**

The Model node provides the robot model name.  
e.g. R-2000iC/165F

**ProductCode                      String type**

The ProductCode node provides the product code of the robot.  
 Since there is no corresponding data, null is set to the node value.

**SerialNumber                      String type**

The SerialNumber node provides the serial number of the robot.  
 Since there is no corresponding data, null is set to the node value.

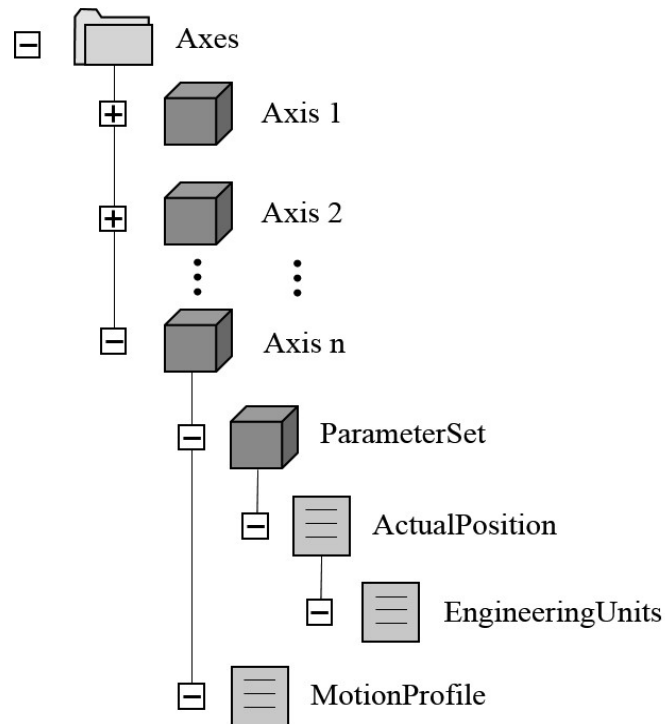
**MotionDeviceCategory                      MotionDeviceCategoryEnumeration type**

The MotionDeviceCategory node provides the category of the robot.  
 The category value corresponding to the following table is set to the node value.

| Category          | Value | Description                                                                                                                                                                                                                                                                                          |
|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OTHER             | 0     | Any motion-device which is not defined.                                                                                                                                                                                                                                                              |
| ARTICULATED_ROBOT | 1     | This robot design features rotary joints and can range from simple two joint structures to 10 or more joints. The arm is connected to the base with a twisting joint. The links in the arm are connected by rotary joints.                                                                           |
| SCARA_ROBOT       | 2     | Robot has two parallel rotary joints to provide compliance in a selected plane.                                                                                                                                                                                                                      |
| CARTESIAN_ROBOT   | 3     | Cartesian robots have three linear joints that use the Cartesian coordinate system (X, Y, and Z). They also may have an attached wrist to allow for rotational movement. The three prismatic joints deliver a linear motion along the axis.                                                          |
| SPHERICAL_ROBOT   | 4     | The arm is connected to the base with a twisting joint and a combination of two rotary joints and one linear joint. The axes form a polar coordinate system and create a spherical-shaped work envelope.                                                                                             |
| PARALLEL_ROBOT    | 5     | These spider-like robots are built from jointed parallelograms connected to a common base. The parallelograms move a single end of arm tooling in a dome-shaped work area.                                                                                                                           |
| CYLINDRICAL_ROBOT | 6     | The robot has at least one rotary joint at the base and at least one prismatic joint to connect the links. The rotary joint uses a rotational motion along the joint axis, while the prismatic joint moves in a linear motion. Cylindrical robots operate within a cylindrical-shaped work envelope. |

**Axes**

The Axes node provides the axis information.  
 It has as many Axis nodes as the number of robot axes. This supports until 9 axes for extended axis and independent axis.



### **ActualPosition      Double type (8 byte)**

The ActualPosition node provides the current position of the axis.

### **EngineeringUnits      EUInformation type**

The EngineeringUnits node provides the units for the ActualPosition node.

The unit of ActualPosition is “deg.” for rotary axis and “mm” for linear axis.

### **MotionProfile      MotionProfileEnumeration type**

The MotionProfile node provides whether the axis is linear or rotary and whether it rotates continuously.

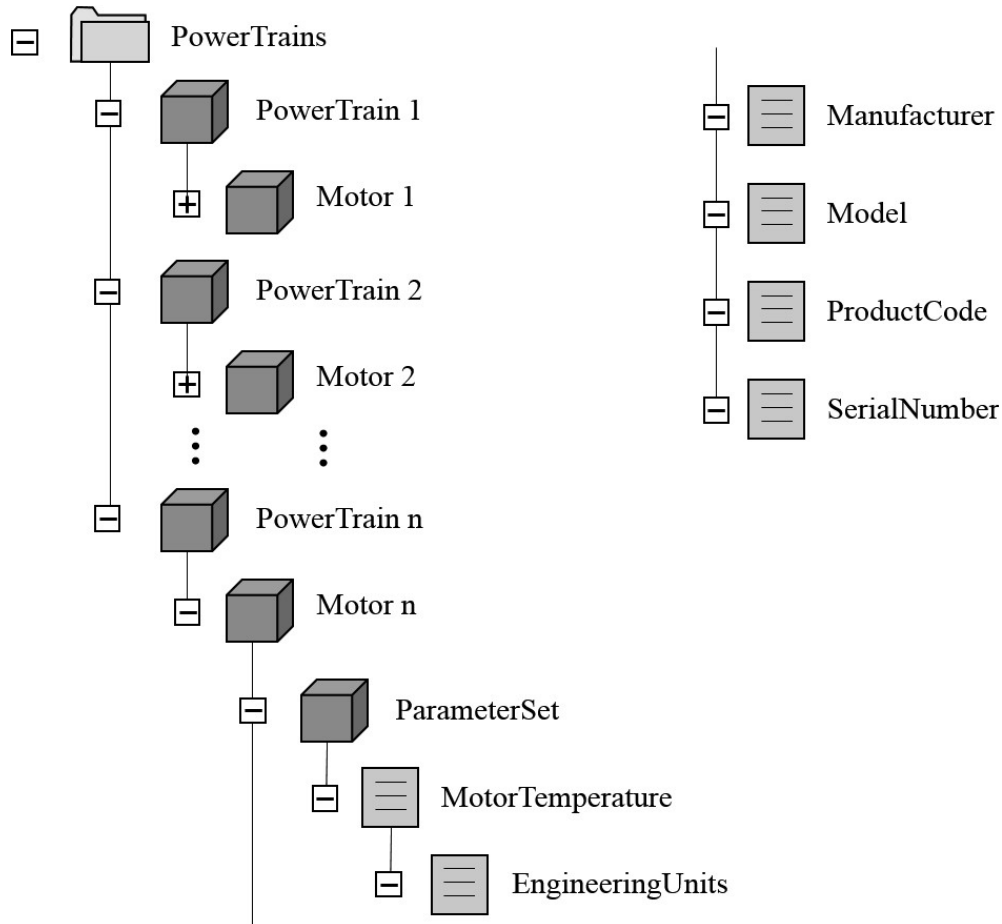
The category value corresponding to the following table is set to the node value.

| Category       | Value | Description                                          |
|----------------|-------|------------------------------------------------------|
| OTHER          | 0     | The axis that does not fit into any category.        |
| ROTARY         | 1     | The axis is rotary and does not rotate continuously. |
| ROTARY_ENDLESS | 2     | The axis is rotary and rotates continuously.         |
| LINEAR         | 3     | The axis is linear and does not rotate continuously. |
| LINEAR_ENDLESS | 4     | The axis is linear and rotates continuously.         |

### **PowerTrains**

The PowerTrains node provides motor information.

It has as many PowerTrain nodes as the number of robot axes and as many Motor nodes as the number of motor. This supports until 9 axes for extended axis and independent axis.

**MotorTemperature      Double type (8 byte)**

The MotorTemperature node provides the motor temperature.  
 Since there is no corresponding data, null is set to the node value.

**EngineeringUnit      EUInformation type**

The EngineeringUnit node provides the units for the MotorTemperature node.  
 Since there is no corresponding data, null is set to the node value.

**Manufacturer      LocalizedText type**

The Manufacturer node provides the name of the motor manufacturer.  
 "FANUC" is set to the node value.

**Model      LocalizedText type**

The Model node provides the motor model name.  
 e.g. aiSR22/4000 80A

**ProductCode      String type**

The ProductCode node provides the product code of the motor.  
 Since there is no corresponding data, null is set to the node value.

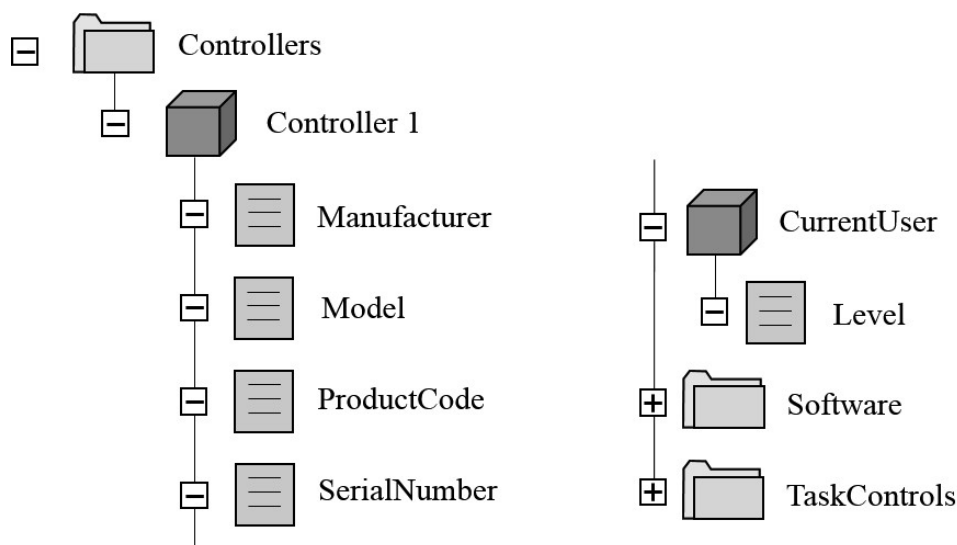
**SerialNumber      String type**

The SerialNumber node provides the serial number of the motor.  
 Since there is no corresponding data, null is set to the node value.



### 55.5.2.2 Controllers

The Controllers node provides information on the model, software, and tasks of the robot controller. It has Controller 1 node. Since the robot controller becomes the OPCUA server, there is only one Controller node.



#### **Manufacturer**      **LocalizedText type**

The Manufacturer node provides the name of the robot controller manufacturer.  
“FANUC” is set to the node value.

#### **Model**      **LocalizedText type**

The Model node provides the robot controller model name.  
e.g. R-30iB Plus

#### **ProductCode**      **String type**

The ProductCode node provides the product code of the robot controller.

#### **SerialNumber**      **String type**

The SerialNumber node provides the serial number of the robot controller.

#### **Level**      **String type**

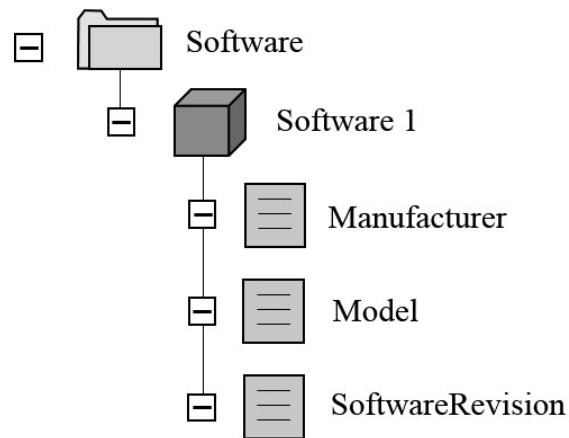
The Level node provides the level of the logged-in user.  
The user level corresponding to the following table is set to the node value.

| Value    | User Level |
|----------|------------|
| INSTALL  | Install    |
| SETUP    | Setup      |
| PROGRAM  | Program    |
| OPERATOR | Operator   |

For the operation authority of each level, refer to the “Operator’s Manual (Basic Function)” (B-83284EN) “9.11 Password Function”.

#### **Software**

The Software node provides information on the software installed on the robot controller.  
The Software node has the Software1 node.



### **Manufacturer**      **LocalizedText type**

The Manufacturer node provides the name of the software manufacturer.  
 "FANUC" is set to the node value.

### **Model**      **LocalizedText type**

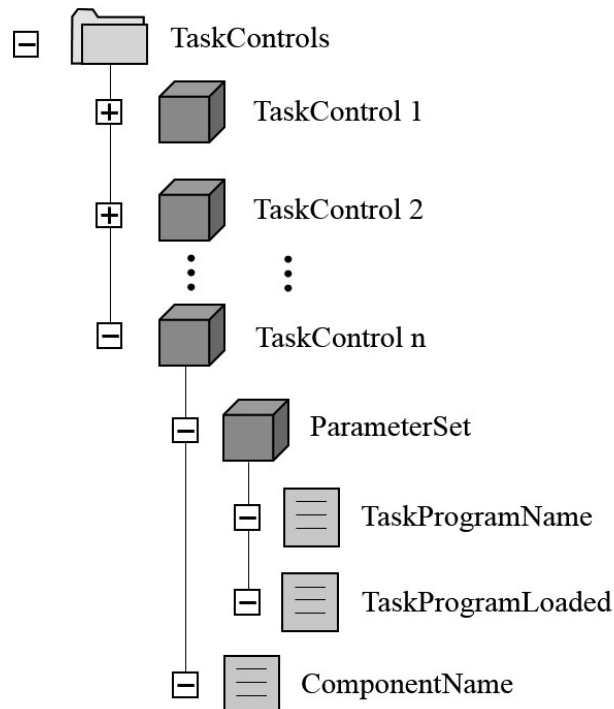
The Model node provides the software model name.  
 e.g. R-30iB Plus

### **SoftwareRevision**      **String type**

The SoftwareRevision node provides the version of the software.  
 e.g. 7DF1/19 (V9.10P/19)

## **TaskControls**

The TaskControls node provides information on user tasks.  
 It has as many TaskControl nodes as the number of tasks set in the maximum number setting.



**TaskProgramName String type**

The TaskProgramName node provides the name of the user program that is running or has finished running.

If there is no corresponding program, null is set.

**TaskProgramLoaded Boolean type (1 byte)**

The TaskProgramLoaded node provides the loading status of the user program.

This node is set to 1 if the user program has been loaded and 0 if not.

**ComponentName LocalizedText type**

The ComponentName node provides the name of the user rewritable TaskControl.

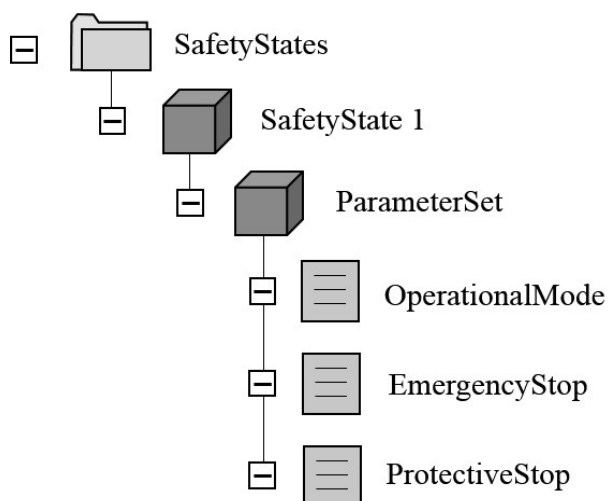
The name of the user program that is running or has finished running is set by default.

The set ComponentName is reset when the power is turned off.

**55.5.2.3 SafetyStates**

The SafetyStates node provides information on operating modes and emergency stop signals.

The SafetyStates node has the SafetyState 1 node.

**OperationalMode OperationalMode type**

The OperationalMode node provides the operation mode of the robot.

The operation mode corresponding to the following table is set to the node value.

| Operation mode | Value |
|----------------|-------|
| OTHER          | 0     |
| T1             | 1     |
| T2             | 2     |
| AUTO (Local)   | 3     |
| AUTO (Remote)  | 4     |

**EmergencyStop Boolean type (1 byte)**

The EmergencyStop node provides the status of the emergency stop signal.

This node is set to 1 when any one of the SOP E-STOP, TP E-STOP, and Ext E-STOP is input, and is set to 0 otherwise.

**ProtectiveStop          Boolean type (1 byte)**

The ProtectiveStop node provides the status of the stop signal.

This node is set to 1 when any one of the Fence Open, TP Deadman, TP Enable, Hand Broken, Overtravel, Low Air Alarm, Belt Broken, SVOFF Input, and Non Teacher Enb.Dev. is input, and is set to 0 otherwise.